

Mathematical Models of Pollinator Populations: A Multi-Scale Review and Research Agenda for Honey Bees and Beyond

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Abstract

Managed and wild pollinator populations have declined at rates far exceeding historical baselines for more than three decades. The decline is driven by an interacting web of stressors — the ectoparasitic mite *Varroa destructor*, multiple viral and microsporidian pathogens, nutritional deficits arising from landscape simplification, agrochemical exposure, climate variability and thermal extremes, and deteriorating queen quality — none of which, acting alone, accounts for observed patterns of colony failure. Yet the great majority of existing mathematical models still treat these stressors in isolation. This review synthesizes the state of mathematical modelling for pollinator populations, with honey bees (*Apis mellifera*) as the analytically best-developed case and with explicit attention to bumble bees, solitary bees, and plant–pollinator communities. We organize the literature around five thematic pillars identified at an American Institute of Mathematics (AIM) workshop convened in March–April 2026: (i) colony persistence, tipping points, and collapse; (ii) parasites and pathogens; (iii) climate and agrochemical stress; (iv) optimization of management interventions; and (v) spatial, landscape, and multispecies dynamics. Across these pillars we trace a common mathematical architecture: a cooperative, non-monotone recruitment term generating an Allee effect, coupled to stressor dynamics that deform the resulting bistable structure until saddle-node or subcritical Hopf bifurcations eliminate the viable-colony attractor. We argue that colony collapse is most rigorously understood as a stochastically driven transition between attractors, and that the distance from a colony’s state to the boundary of the extinction basin — rather than any single stressor level — is the quantity that governs vulnerability. We inventory post-2022 contributions, identify where the literature is thin (multi-pathogen coinfection, formal optimization as opposed to scenario evaluation, coupled within-hive and landscape scales, wild-bee models), and set out sixteen open mathematical problems together with a staged research agenda spanning data infrastructure, pilot studies, and translational tools. Our central methodological recommendation is staged model complexity: begin with analytically tractable systems, extract structural results, and add mechanism only where the simpler model demonstrably fails.

Keywords: pollinator decline; honey bee colony dynamics; *Varroa destructor* destructor; Allee effect; bifurcation and tipping points; optimal control; early warning signals; plant–pollinator mutualism; hybrid mechanistic–machine learning modelling.

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1 Introduction

1.1 The problem and why it is mathematical

The western honey bee, *Apis mellifera*, is the most economically consequential managed pollinator in the world. Animal pollination is required or strongly beneficial for roughly three-quarters of the leading global food crops [Klein et al., 2007], and managed honey bees alone contribute billions of dollars annually in pollination services to United States agriculture [Morse and Calderone, 2000]. The ecological reach of pollinators extends well beyond agriculture: bees are keystone mutualists in natural plant communities across temperate and subtropical regions, sustaining the reproductive success of wildflowers on which a large fraction of terrestrial invertebrate and vertebrate diversity depends [Bascompte and Jordano, 2007].

Against this backdrop, managed colony losses in the United States have persistently exceeded 30–50% per year, far above the roughly 10% that beekeepers historically regarded as sustainable [Bruckner et al., 2023, Steinhauer et al., 2021]. The appearance of Colony Collapse Disorder (CCD) in the mid-2000s — the rapid disappearance of adult workers from otherwise provisioned, queen-right hives — focused scientific and policy attention on the problem [vanEngelsdorp et al., 2008]. Two decades of subsequent research have made clear that CCD is not a single disease but the phenotypic expression of a multi-factorial system failure. The risk factors most consistently identified in epidemiological studies — *Varroa destructor* infestation, co-occurring viral infections (particularly deformed wing virus, DWV, and acute bee paralysis virus, ABPV), nutritional stress from landscape degradation, sublethal pesticide exposure, and climate variability — are individually insufficient to explain observed collapse patterns. Their *interactions* appear to be necessary [Dainat et al., 2012, Goulson et al., 2015].

This is precisely the structure of problem for which mathematics is indispensable. How stressors amplify one another, generate threshold effects, produce abrupt and irreversible transitions, and respond to management interventions are questions about the geometry of a nonlinear dynamical system. They are not accessible to correlational or descriptive approaches alone. Bifurcation theory characterizes tipping points; persistence theory distinguishes coexistence from extinction; optimal control converts management heuristics into state-dependent decision rules; stochastic analysis quantifies the probability and timing of collapse under environmental noise; and modern statistical and machine-learning methods connect all of these to the sparse, heterogeneous data that pollinator monitoring actually produces.

1.2 Scope, and what distinguishes this review

Mathematical modelling of honey bee populations was reviewed comprehensively by Chen et al. [2021], who catalogued colony demography models, *Varroa destructor* and virus models, and the then-current state of seasonality and age structure. The present review differs in three respects.

- (i) **Scope beyond the honey bee.** We treat *Apis mellifera* as the analytically best-developed case rather than as the object of study. Bumble bee colony models, solitary and wild bee landscape models, and plant–pollinator community models are included where they bear on the same mathematical questions. This matters because several of the most important open problems — phenological mismatch, landscape-mediated competition, mutualism collapse — are not honey-bee problems at all.
- (ii) **Organization by mathematical question rather than by stressor.** We show that a small number of recurring structures — non-monotone recruitment producing Allee effects, cooperative (monotone) system structure, delay-induced destabilization, and periodic or stochastic parameter forcing — underlie results that are usually presented as disconnected.
- (iii) **An explicit forward agenda.** Roughly one third of this article is devoted to open problems and to a staged research programme, reflecting the purpose for which it was written: to consolidate the outcomes of the AIM workshop [AIM Workshop Participants, 2026] into a shared agenda for a community that is currently fragmented across disciplines and institutions.

The material synthesized here draws on the workshop’s eight working groups, which spanned mathematical biology, dynamical systems, ecology, entomology, virology, data science, geospatial analysis, and commercial beekeeping.

It is worth recording that concern about pollinator decline is not new. The entomologist Edith Patch warned in the 1930s of declining bee populations driven by agricultural intensification, reduced forage diversity, and early pesticide use [Patch, 1937] — a diagnosis that names three of the seven stressor categories in Table 1. The mathematical treatment of the problem is far younger, and the mismatch between the urgency of the ecology and the maturity of the mathematics deployed against it is part of what this review is written to address.

1.3 Organization

Section 2 summarizes the biological substrate and the seven-stressor framing that recurs throughout. Section 3 traces the modelling lineage from simulation platforms to analytically tractable ordinary and delay differential equation (ODE/DDE) systems. Sections 4–8 treat the five thematic pillars in turn, each closing with open problems. Section 9 covers the cross-cutting methodology of data, inference, and hybrid mechanistic–machine-learning modelling. Section 10 consolidates the structural gaps and the full open-problem list, and Section 11 sets out the research agenda. Section 12 concludes. Appendix A provides a structured inventory of post-2022 contributions by theme.

2 Biological substrate and the multi-stressor problem

2.1 The colony as a superorganism

A honey bee colony is a tightly integrated social unit of roughly 20,000–60,000 individuals in which division of labour, communication, and collective decision-making generate behaviour that is not predictable from individual bee physiology [Seeley, 2010]. A healthy colony comprises a single laying queen, thousands of female workers at various developmental and behavioural stages, and a seasonal cohort of drones. Workers pass through a fixed developmental sequence — eggs hatch into larvae nursed for approximately nine days, then are capped as pupae for roughly twelve days before eclosion — giving a developmental delay of about 21 days. This delay is close to a biological constant, and it has consequences for both population dynamics and disease transmission that no delay-free model reproduces exactly.

Adult workers progress from in-hive tasks (cell cleaning, brood care, food processing, construction, defence) to foraging. The transition is mediated by age but modulated by colony state: it is governed in part by vitellogenin, whose titre declines with age [Amdam et al., 2006, Rodriguez Messan et al., 2018], and by inhibitory social signalling from existing foragers. Perry et al. [2015] demonstrated that nutritional stress accelerates this behavioural maturation, causing workers to forage before their physiology optimally supports it. Precocious foragers are less efficient and shorter-lived, which depletes the in-hive nursing population and accelerates further precocious foraging — a positive feedback that can drive an apparently healthy colony to failure. This mechanism is the biological origin of most of the Allee-type structure discussed in Section 4.

Three features of this biology are load-bearing for everything that follows: colony function is *cooperative* (per-capita performance improves with population size over a range), *delayed* (today’s adults were provisioned weeks ago), and *seasonal* (laying rate, forage availability, and thermal demand all vary periodically).

2.2 *Varroa destructor destructor*: biology that dictates model structure

Varroa destructor is an ectoparasitic mite that feeds on the fat body of developing and adult bees. Its host shift from *Apis cerana* to *Apis mellifera* in the mid-twentieth century is among the most consequential

host–parasite range expansions in agricultural history [Rosenkranz et al., 2010]. The life cycle has two phases with distinct mathematical properties:

- the *phoretic* phase, in which adult female mites ride adult bees and disperse within and between colonies through forager exchange, functioning principally as dispersal agents and disease vectors;
- the *reproductive* phase, in which foundress mites enter brood cells before capping and reproduce inside the sealed cell, producing one to three viable female offspring per cycle, invisible to the beekeeper and shielded from most topical treatments.

Two consequences follow directly. First, the observable phoretic mite load is a biased and lagged estimate of total mite abundance, so any control rule stated in terms of phoretic counts is a rule stated in terms of a partially observed state. Second, the two phases give the bee–mite interaction an intrinsically stage-structured, time-delayed character. Both facts argue for state-dependent, model-guided treatment timing rather than fixed thresholds (Section 7).

Varroa’s most damaging effect is arguably not parasitism but vectored transmission. The mite is the principal vector for DWV and several other RNA viruses, elevating viral titres in infested colonies by orders of magnitude and driving the global emergence of virulent DWV variants [Wilfert et al., 2016, Posada-Flórez et al., 2024]. The interaction is bidirectional: infestation facilitates transmission, while infection alters host physiology and may alter mite reproductive success. The result is a synergistic parasite–pathogen complex with qualitatively new dynamical properties relative to either stressor alone (Section 5).

2.3 Pathogens beyond *Varroa destructor* and DWV

More than twenty RNA viruses infect honey bees, of which only a small subset have been modelled [Brutscher et al., 2015]. DWV is the most prevalent and best studied, and exists in strains (DWV-A, DWV-B) with differing virulence. ABPV, sacbrood virus, black queen cell virus, and chronic bee paralysis virus all cause significant field pathology. Colonies routinely carry several viruses simultaneously, yet the competitive, synergistic, or neutral character of these interactions is largely uncharacterized. Microsporidian parasites (*Nosema/Vairimorpha* spp.) constitute a second, distinct disease axis, and are increasingly co-monitored with *Varroa destructor* in national surveillance [Abban et al., 2024].

Transmission occurs by two routes with different consequences for model structure. *Horizontal* transmission proceeds through direct contact, trophallaxis, and mite vectoring, and is typically frequency-dependent. *Vertical* transmission passes virus from infected mothers to offspring and from reproductively active infected mites to their progeny; it is density-dependent and sustains a persistent reservoir. Models combining both routes with explicit *Varroa destructor* life-cycle structure generate substantially richer dynamics than single-route models, including disease persistence at low mite densities [Kang et al., 2016].

2.4 Nutrition, agrochemicals, and the remaining stressors

Beyond parasites and pathogens, five further stressor categories are implicated consistently in the empirical record. *Nutritional stress* operates through pollen diversity, protein quality, and the availability of essential phytonutrients — amino acids, lipids, sterols, phenolics — all of which affect longevity, immune competence, and reproductive success [Di Pasquale et al., 2013]. Landscape simplification has reduced pollen nutritional quality across many regions [Donkersley et al., 2014], and Moore et al. [2025] demonstrated that specific sterols absent from conventional pollen substitutes are required to sustain brood production. *Agrochemical exposure* occurs through hundreds of compounds encountered in foraging and applied within hives, with well-documented sublethal effects at environmentally realistic concentrations [Blacqui re et al., 2012]. *Climate variability* affects every phase of colony biology, from egg-laying rate to thermoregulatory demand. *Cumulative forager flight hours* are bounded by wing wear and metabolic depletion, so sparse landscapes requiring long foraging trips can deplete the adult population faster than brood matures to replace it [Perry et al., 2015]. Finally, *queen quality* is both cause and consequence of

colony stress: mite-vectoring viruses compromise queen health, while a failing queen triggers colony-level responses that may further reduce viability [vanEngelsdorp and Meixner, 2010].

2.5 The seven-stressor framing

The workshop consolidated the empirical literature into seven stressor categories, listed in Table 1 together with the modelling handle each currently admits. Two observations are worth recording. First, the categories differ enormously in how well they are represented mathematically: *Varroa destructor* and viruses are comparatively mature, while queen quality and cumulative forager flight hours are barely represented at all. Second, and more importantly, the empirical evidence points to *interactions* among these categories as the driver of the transition from manageable stress to irreversible collapse. Formalizing and quantifying those interactions is the central mathematical challenge of the field.

Table 1: The seven stressor categories identified at the AIM workshop, with the mathematical representation each currently admits and an assessment of modelling maturity.

Stressor	Typical mathematical representation	Maturity
<i>Varroa destructor</i> mites	Coupled state variable with phoretic/reproductive stages; parasitism raises effective adult mortality and reduces brood survival	Mature
Viral pathogens	SI or SIR compartments on bees and mites; frequency-dependent horizontal plus density-dependent vertical transmission	Mature (single virus)
Nutrition and forage	Food or nutrient state variable modulating laying rate and longevity; landscape-derived intake function	Developing
Climate and thermal extremes	Periodic or stochastic forcing of vital rates; thermoregulation as a colony-size-dependent feedback	Developing
Agrochemicals	Exposure–dose–effect pathway mapping residue concentration to cohort-specific mortality or performance loss	Developing
Forager flight hours	Cumulative-damage or age-of-onset variable coupled to social inhibition	Sparse
Queen quality	Time-varying laying rate; rarely an explicit dynamic variable	Sparse

3 Modelling foundations

3.1 Simulation and agent-based platforms

The earliest quantitative colony models were simulation-based and developed largely by colony biologists. DeGrandi-Hoffman et al. [1989] pioneered the BEEPOP family (later VarroaPop and BeePop+), which tracks brood, adults, and mite load on a daily or weekly step using empirically measured stage-transition rates and environmental forcing for weather and forage. Martin [1998] developed simulation models focused specifically on within-colony *Varroa destructor* population growth and its consequences for the timing of collapse, and Wilkinson and Smith [2002] extended this line to mite immigration and reinvasion. BEEHAVE [Becher et al., 2014] added an explicit agent-based foraging module coupled to a landscape representation, and Bumble-BEEHAVE [Becher et al., 2018] carried the same architecture to bumble bee colony, population, and community levels.

These platforms have been practically influential and underpin much current management guidance. Their limitation is analytical: they do not yield closed-form predictions, and they do not readily support the identification of thresholds, bifurcation structure, or optimal control laws. They are best regarded as high-fidelity numerical experiments that a tractable model should be able to explain.

3.2 Compartmental ODE models of colony demography

Analytically tractable colony models begin with Khoury et al. [2011], who formulated a two-compartment system for hive bees H and foragers F with a social-inhibition recruitment term:

$$\frac{dH}{dt} = L - HR(H, F), \quad \frac{dF}{dt} = HR(H, F) - mF, \quad (1)$$

where L is the eclosion rate, m forager mortality, and R a recruitment function decreasing in $F/(H + F)$, so that a shortage of foragers accelerates recruitment. Khoury et al. [2013] extended the framework to include stored food, and Perry et al. [2015] added cumulative flight experience and nutritional state, formalizing the precocious-foraging feedback described above. Russell et al. [2013] and Chen et al. [2020] developed stage-structured variants and tested alternative brood-to-adult formulations against colony census data; the latter found that a formulation in which adults arise from surviving eggs, combined with seasonal queen laying, fits observed dynamics better than simpler brood-to-adult transitions.

The essential structural result of this family is that colonies possess a *minimum viable population*: below a critical adult number the recruitment feedback cannot sustain the colony, and it collapses regardless of external resources. This is the Allee effect analysed in Section 4.

A widely used reduced form replaces the two-compartment structure with a single adult variable H and a non-monotone per-capita recruitment term,

$$\frac{dH}{dt} = \frac{rH^2}{K + H^2} - d_h H, \quad (2)$$

where r is the maximum eclosion rate (proportional to queen laying capacity), K is an Allee parameter, and d_h is adult mortality. Despite its simplicity, (2) reproduces the qualitative bistability of the full demographic models and serves as the base layer on which stressor modules are built throughout this review.

3.3 Age structure and delay differential equations

The fixed developmental delays of bee biology motivate explicit delay formulations. Writing τ for the egg-to-adult delay and $\theta(t)$ for stressor-dependent mortality during development, brood dynamics take the form

$$\frac{dB}{dt} = \frac{rH(t)^2}{K + H(t)^2} - \exp\left(-\int_{t-\tau}^t \theta(s) ds\right) \frac{rH(t-\tau)^2}{K + H(t-\tau)^2} - \theta(t)B, \quad (3)$$

in which the exponential factor is the survival probability through the capped stage [Messan et al., 2017, 2021]. A structurally important result is that the qualitative stability architecture of the delay-free system is largely preserved under the introduction of delays: the Allee effect and bistability persist, and the delay primarily relocates equilibria rather than altering their stability type. Chen et al. [2023] extended this to seasonal parasitism and showed that seasonal forcing combined with parasitism can nonetheless destabilize dynamics that would be stable under either alone.

This result licenses a research strategy that recurs throughout this review: *use tractable ODE systems to extract structure, and reserve DDE and age-structured formulations for quantitative prediction and model fitting*. It is a licence, not a blanket permission — delays do matter for oscillation onset and for the timing of interventions — but it explains why so much can be learned from low-dimensional systems.

3.4 A taxonomy of model classes

Table 2 organizes the principal model classes by what each is good for. The recurring lesson from the workshop is that these classes are complementary rather than competing, and that the field's weakness is not any individual class but the absence of systematic translation between them.

Table 2: Model classes for pollinator population dynamics, with characteristic strengths and limitations.

Class	Strengths	Limitations
Simulation / agent-based	High biological fidelity; direct use of measured rates; landscape and weather coupling; regulatory acceptance	No closed-form results; thresholds only found by scanning; many parameters, weak identifiability
Low-dimensional ODE	Bifurcation and stability analysis; explicit persistence criteria; basis for optimal control	Omits age structure and delays; aggregates heterogeneity
DDE / age-structured	Correct developmental timing; oscillation onset; realistic fitting to census data	Harder analysis; stability results often numerical
PDE / spatial	Within-hive thermal structure; forager redistribution over landscapes	Existence and stability theory incomplete for coupled ODE–PDE systems
Stochastic (SDE, jump)	First-passage times to extinction; effect of extreme events; variance-based early warning	Parameterizing noise is difficult; few honey bee applications to date
Statistical / ML	Extracts signal from sparse, heterogeneous sensor data; detection and classification	Limited extrapolation; interpretability; scarce labelled collapse events

4 Theme I: Persistence, tipping points, and collapse

Colony persistence depends on the interaction of colony demographics with biological and environmental stressors. Mathematical models help distinguish gradual declines in colony performance from abrupt changes in outcome, such as the loss of a viable equilibrium or movement across the boundary between persistence and extinction. This theme examines how Allee effects, thermoregulation, parasitism, and bistability influence persistence, followed by a discussion of recent work on seasonal variation and interacting stressors. A central limitation of the current literature is that few models examine several stressors together within a single mathematical framework.

4.1 The Allee effect and its mechanistic origins

The Allee effect — reduced per-capita growth at low density — is a defining feature of social bee colonies, and unusually, one with a transparent mechanistic basis [Allee, 1931]. Below a critical mass of adult workers a colony cannot simultaneously maintain thermoregulation, brood care, foraging, and defence. Its mathematical signature is a recruitment function that is non-monotone in per-capita terms, as in (2).

System (2) admits the extinction equilibrium $H = 0$ together with two positive equilibria whenever

$$r > 2d_h\sqrt{K}, \quad (4)$$

namely an unstable lower equilibrium H_c^* (the Allee threshold) and a stable upper equilibrium H_s^* (the viable colony). When (4) fails — because nutritional stress or queen deterioration reduces r , or because parasitism or thermal stress raises d_h — the two positive equilibria annihilate in a saddle-node bifurcation and extinction becomes globally attracting. This is the simplest rigorous statement of a fact that the empirical literature reports repeatedly: nutritional or thermal stress alone, with no pathogen involvement whatsoever, is sufficient to collapse a colony.

It is worth emphasizing how many distinct biological mechanisms funnel into this same structure. Perry et al. [2015] obtain it from precocious foraging; Khoury et al. [2013] from food storage feedback; Zeaiter and Myerscough [2020] from thermoregulation, as discussed below. That several mechanisms produce the same normal form is theoretically satisfying, but it is also a warning: the presence of an Allee effect in data does not identify which mechanism generated it, and distinguishing them requires either mechanism-specific covariates or manipulative experiments.

4.2 Thermoregulatory Allee effects

Zeaiter and Myerscough [2020] gives a particularly clean example. They extend the Perry–Barron framework with a delay differential equation model of stored food, uncapped brood, hive bees, and foragers, in which thermoregulatory capacity is represented not by an explicit temperature variable but by hive bee abundance. Crucially, they separate the *current* energetic cost of thermoregulation from its *delayed* developmental consequences: current hive bees determine present food consumption, whereas the historical hive bee population present when today’s foragers were capped brood determines those foragers’ quality. The stress function takes the form

$$S(H_{\tau_d}) = \frac{H_{\tau_d}^2 + \psi\beta^2}{H_{\tau_d}^2 + \beta^2}, \quad (5)$$

where H_{τ_d} is the average hive bee population during the capped stage of the current forager cohort, ψ is the maximum multiplicative effect on mortality, and β is the half-saturation population.

The resulting feedback — poor thermoregulation produces low-quality foragers, which collect less food and die sooner, which reduces the hive bee population, which further degrades thermoregulation — generates bistability and collapse through saddle-node bifurcation. Two findings deserve emphasis. First, colony outcome is determined principally by demographic structure and the thermoregulatory feedback rather than by food quantity: supplemental feeding delays collapse but does not prevent it. Second, the effect is produced entirely without an explicit temperature variable, which is simultaneously the model’s elegance and its principal limitation. Temperature does not appear, so heat waves, cold snaps, hive insulation, and seasonal variation cannot be represented, and the stress function is phenomenological rather than fitted to field thermal data.

4.3 Parasitism and the catastrophe event

Coupling mites to the base colony model changes the bifurcation structure qualitatively. Let M denote mite abundance, α the parasitism rate, c mite reproductive efficiency, and d_m mite mortality. The mite nullcline fixes the adult bee density at which mite growth balances mite death,

$$H^* = \frac{d_m}{c\alpha}, \quad (6)$$

and the position of H^* relative to $\sqrt{K}$ determines the qualitative outcome. Applying persistence and bistability results in the spirit of Thieme [1993], Thieme et al. [2009]:

- If $H^* > \sqrt{K}$, the interior equilibrium (H^*, M^*) is a stable sink: bees and mites coexist at positive densities, and the colony survives in a compromised but persistent state.
- If $H^* < \sqrt{K}$, the interior equilibrium loses stability through a *subcritical* Hopf bifurcation. Unstable periodic orbits emerge and expand until they collide with the stable manifold of the boundary saddle $(H_c, 0)$. Beyond that collision, trajectories that would otherwise remain in the viable basin are routed to extinction, and bees and mites collapse together.

We refer to the second case as a *catastrophe event*, with the criterion

$$H^* = \frac{d_m}{c\alpha} < \sqrt{K}. \quad (7)$$

It is a rigorous dynamical analogue of CCD: collapse is abrupt, effectively irreversible, and — critically — can occur in colonies whose mite loads had appeared manageable. Criterion (7) also has an immediate management reading. Anything that raises the Allee parameter K (malnutrition, disease, queen age), increases mite reproductive efficiency c or parasitism rate α (susceptible genetics, abundant brood) or lowers effective mite mortality d_m (acaricide resistance) moves the system toward the catastrophe regime. Stressors are not additive in their effect on colony survival; they combine multiplicatively within a single scalar criterion.

4.4 Bistability, path dependence, and the composite stress score

In the bistable regime, the outcome depends not only on current stressor levels but on the trajectory by which the colony arrived at its current state. Two colonies with identical mite loads and nutritional status in October may have entirely different collapse probabilities in December, because one is inside the viable basin and the other outside it. Standard threshold protocols — treat when phoretic mite load exceeds a fixed percentage — are structurally blind to this path dependence, which is one mathematical explanation for the persistent observation that colonies within a single apiary, with similar queens, similar treatments, and shared forage, nonetheless diverge dramatically in outcome.

The workshop proposed aggregating stressor effects into a composite stress score Σ , defined operationally as a weighted combination of standardized stressor indices and interpreted geometrically as a signed distance from the bifurcation boundary in parameter space. Colonies with Σ above a calibrated threshold lie in the viable basin; those below are at elevated risk. The appeal of Σ is that it converts a high-dimensional monitoring problem into a one-dimensional risk statistic with a dynamical interpretation. The difficulty is that estimating it requires knowing the basin boundary, which is exactly what is not known. This tension defines the first open problem below. A complementary approach was raised at the workshop by analogy with soil health assessment, where a large panel of indicators is reduced to a scored index through a documented aggregation protocol [Nunes et al., 2021]; whether such an index can be given a dynamical interpretation is an open question in itself.

4.5 Post-2022 developments

Recent contributions have broadened the persistence literature in three directions. Seasonality and non-autonomous formulations now dominate: Ibrahim and Dénes [2022] model parallel spread of mites and vectored virus with time-periodic parameters, Chen et al. [2023] analyse seasonality combined with parasitism, and Alharbi [2024] derive reproduction numbers in both autonomous and non-autonomous settings for interacting colonies. Threshold dynamics with general eclosion and recruitment rates have been formulated in terms of an explicit basic reproduction number determining survival or collapse [[Authors to verify], 2025]. Climate-driven simulation has entered the persistence literature through Rajagopalan et al. [2024], discussed in Section 6. General colony population and honey production models continue to appear [Patiño-Romero et al., 2022]. Appendix A, Table 5, inventories this literature.

What remains conspicuously absent is any analysis treating more than two stressor categories simultaneously as a coupled dynamical system.

Open Problem 4.1 (Basin boundary characterization). Characterize the basins of attraction of the viable-colony and extinction equilibria as explicit functions of the seven stressor parameters of Table 1. Derive computable estimates of the separatrix from the quantities actually available in colony monitoring — frame counts, mite washes, brood-to-adult ratio, stored food — so that the composite stress score Σ can be evaluated in the field rather than only in parameter space.

Open Problem 4.2 (Mechanism identification). Several distinct biological mechanisms (precocious foraging, food storage feedback, thermoregulatory degradation) generate the same saddle-node normal form. Determine what observations — and at what sampling frequency — suffice to identify which mechanism is operating in a given colony, or prove that they are non-identifiable from standard monitoring data.

Recent studies have extended persistence models to include seasonal variation, interacting colonies, and climate-driven changes. However, most still focus on a single stressor or on a limited number of stressors considered in pairs. We therefore lack a general framework for determining how combined stressors alter the conditions for persistence, the threshold for extinction, and the possibility of recovery.

5 Theme II: Parasites, pathogens, and the coinfection gap

Parasites and pathogens affect colony dynamics through host mortality, changes in adult and brood populations, and transmission among bees and mites. Mathematical models can identify invasion thresholds, disease free and infected equilibria, and conditions under which infection alters colony persistence. This theme reviews models that couple bee, mite, and virus populations; examines reproduction numbers for infection invasion and persistence; and analyzes how these interactions affect equilibrium stability and long term colony outcomes. It then considers recent work beyond *Varroa destructor* and DWV and identifies a major gap in the literature: most models include either one pathogen or one pathogen and vector pair, while few represent simultaneous infection by multiple pathogens together with nutritional, pesticide, or thermal stress.

5.1 Coupled bee–mite–virus systems

Early models treated mite load as an exogenous constant [Sumpter and Martin, 2004] or assumed fixed mite-to-bee ratios [Eberl et al., 2010], which permits analysis of infestation *level* but not of the coupled dynamic. Ratti et al. [2013, 2015] modelled honey bee–*Varroa destructor*–ABPV co-dynamics with seasonal (time-periodic) coefficients and used Floquet theory to analyse the stability of virus-free periodic solutions; Ratti et al. [2017] extended the framework to forager loss. Bernardi and Venturino [2016] analysed an *SI–SI* system for adult bees and mites, characterizing the full set of outcomes: healthy-bee persistence, endemic disease with mite elimination, *Varroa destructor* invasion with extinction of healthy bees, and coexistence. Related epidemic formulations include Betti et al. [2014] and Kribs-Zaleta and Mitchell [2014], the latter treating CCD itself as a contagion process.

The first fully coupled bee–mite–virus ODE model incorporating both mite phases together with horizontal and vertical viral transmission was formulated by Kang et al. [2016]. In condensed form, with S_h, I_h susceptible and infected adult bees, S_m, I_m susceptible and infected phoretic mites, $H = S_h + I_h$ and $M = S_m + I_m$,

$$\frac{dS_h}{dt} = \frac{r(S_h + \rho I_h)^2}{K + (S_h + \rho I_h)^2} - d_h S_h - \Lambda_h(S_h, I_h, I_m) - \alpha S_h M, \quad (8)$$

$$\frac{dI_h}{dt} = \Lambda_h(S_h, I_h, I_m) - \alpha I_h M - (d_h + \mu_h) I_h, \quad (9)$$

$$\frac{dS_m}{dt} = S_m \left[\alpha c S_h - \beta_{hm} \frac{I_h}{H} - d_m \right], \quad (10)$$

$$\frac{dI_m}{dt} = c \alpha (I_h M + S_h I_m) + \beta_{hm} \frac{S_m I_h}{H} - (d_m + \mu_m) I_m, \quad (11)$$

where Λ_h collects horizontal and mite-mediated transmission into bees, $\rho \in [0, 1]$ scales the reduced reproductive contribution of infected bees, and μ_h, μ_m are disease-induced mortalities. System (8)–(11) admits up to four equilibria and exhibits the bistability, multiple attractors, and catastrophic transitions of Section 4. Messan et al. [2021] embedded the same interaction in an age-structured, seasonally forced setting.

5.2 Reproduction numbers and why the classical one is insufficient

For the virus-only subsystem ($S_m = I_m = 0$), the disease-free equilibrium is locally stable when

$$\mathcal{R}_0^{\text{bee}} = \frac{\beta_h}{d_h + \mu_h} < 1. \quad (12)$$

With mites present, however, the effective reproduction number acquires contributions from the mite reservoir,

$$\mathcal{R}_0^2 = \frac{\beta_h + \hat{\beta}_{mh} \frac{I_m^*}{M^*} + \tilde{\beta}_{mh} \frac{I_m^*}{H^*}}{d_h + \mu_h}, \quad (13)$$

which depends on equilibrium mite density and on the infected fraction of the mite population. The consequence is structural: it is entirely possible for $\mathcal{R}_0^{\text{bee}} < 1 < \mathcal{R}_0$, so that disease persists and causes damage in a regime where a decoupled bee-only analysis would predict its elimination. Reproduction numbers computed on bee compartments alone systematically underestimate risk in mite-infested colonies, and comparisons between studies that use different definitions of $\mathcal{R}_0$ are correspondingly hazardous.

5.3 Multi-attractor dynamics and parasite-mediated persistence

Numerical analysis of the coupled system in the bistable regime yields a counterintuitive family of outcomes:

- Without virus, mites and bees oscillate and eventually collapse together (criterion (7) is satisfied).
- Without mites, bees and virus coexist at stable positive levels.
- With both, all four populations coexist at elevated stable equilibria.

In the third scenario the virus *stabilizes* the bee–mite system: suppression of mite fitness maintains a larger adult population, which in turn sustains mite growth and viral circulation, producing a form of parasite-mediated persistence. The management implication is uncomfortable but important: aggressive mite treatment in the presence of high viral load can destabilize a system that was in a compromised but stable coexistence state, potentially precipitating the very collapse it was intended to prevent. Protocols that address mite load and viral burden together are, on this analysis, not merely more thorough but qualitatively different in their dynamical consequences. We stress that this is a model-derived prediction that has not been tested experimentally, and that designing such a test is itself a worthwhile deliverable.

5.4 Beyond *Varroa destructor* and DWV: recent expansion

Post-2022 work has broadened the disease agenda in three directions, catalogued in Appendix A, Table 6.

Seasonality. Non-autonomous formulations with time-periodic parameters are now standard rather than exceptional [Ibrahim and Dénes, 2022, Chen et al., 2023, Alharbi, 2024], reflecting a consensus that disease dynamics cannot be separated from colony demography or from the seasonal environment in which it unfolds.

Intervention. Ssenoga et al. [2022] derive mite-free and mite reproduction numbers and compare chemical, mechanical, and traditional control approaches including queen caging and replacement; Umara et al. [2023] and subsequent varroosis models add explicit biocontrol-agent compartments. Schödl et al. [2022] use BEEHAVE to evaluate non-synthetic acaricide regimes including drone brood removal and organic acid treatments.

Microsporidia and multi-colony transmission. Osorio Rodríguez and Toro Zapata [2025] model nosemosis while explicitly incorporating agrochemical exposure and control measures — one of the clearest recent examples of genuine multi-stressor disease modelling. El Hajji et al. [2026] model dual-species *Nosema* transmission across two interacting colonies with distributed delays and an environmental pathogen compartment, and Salman et al. [2026] analyse *Nosema ceranae* dynamics under limited treatment capacity and environmental strain.

5.5 The coinfection gap

Despite this expansion, a specific and consequential gap persists. The overwhelming majority of post-2022 mathematical disease models treat a *single* pathogen or a single pathogen–vector system. Explicit coinfection models involving multiple viruses, or *Varroa destructor*–*Nosema* interaction, or simultaneous disease, nutritional, pesticide, and thermal stress, remain rare. This is not for want of biological motivation: Posada-Flórez et al. [2024] document that colonies routinely harbour several viral species at once, that mite-mediated transmission appears to select for greater virulence, and that the combined effects of mites and viruses are synergistic rather than additive. Large-scale surveillance now routinely co-reports *Varroa destructor* and *Nosema* prevalence [Abban et al., 2024], and empirical coinfection and gut-microbiome studies continue to accumulate. The gap is mathematical, and it is arguably the most tractable high-value target in the disease literature: the modelling tools (multi-strain epidemic theory, competitive exclusion analysis, invasion reproduction numbers) already exist and simply have not been brought to bear.

A second gap is structural rather than topical. Disease models are typically developed *separately* from detailed colony demography, behavioural response, queen dynamics, and apiary-level transmission. The result is a literature in which the epidemiology is sophisticated and the demography is a placeholder, or vice versa, but rarely both.

Open Problem 5.1 (Multi-virus coinfection dynamics). For a colony carrying two viruses with distinct transmission rates, virulences, and possible cross-immunity or interference, characterize the complete set of dynamical outcomes as a function of the relative transmission parameters. Determine whether coinfection generically increases or decreases collapse risk relative to single-virus infection, and identify the parameter regimes in which competitive exclusion between viral strains fails.

Open Problem 5.2 (Cross-taxon parasite interaction). Formulate and analyse a model coupling *Varroa destructor* and *Nosema* dynamics with shared host demography and immune-mediated interaction. Determine whether the two parasites are dynamically synergistic, and whether suppression of one can increase the burden of the other.

Open Problem 5.3 (Treatment-induced destabilization). Derive conditions under which acaricide application moves a colony from a stable coexistence attractor into the extinction basin. Characterize the safe treatment region in the space of (mite load, viral titre, adult population, season), and determine whether staged or partial treatment can avoid the destabilization that abrupt treatment induces.

6 Theme III: Climate, thermoregulation, and agrochemical stress

Climate and agrochemical exposure affect pollinator populations through different mechanisms. Climate models have focused mainly on thermoregulation and seasonal weather, whereas agrochemical models examine how exposure alters survival, behavior, reproduction, and colony dynamics.

6.1 Climate and thermoregulation

Honey bee colonies regulate their internal temperature through collective behavior, but this capacity depends strongly on the colony population and worker activity. Mathematical studies have represented this dependence at different levels, ranging from demographic feedback to explicit spatial temperature dynamics. Zeaiter and Myerscough [2020] incorporated thermoregulation stress into a delay differential equation model of colony demography. Temperature itself is not a state variable in their model; instead, hive-bee abundance determines the degree of thermoregulatory stress. A reduced hive-bee population increases the energetic cost of thermoregulation and produces poorly developed foragers with lower foraging efficiency and shorter lifespans. This feedback generates bistability and an Allee effect, with colony collapse occurring through a saddle-node bifurcation. Thus, colony size can determine whether thermoregulatory stress remains manageable or drives further population decline.

Bastiaansen et al. [2020] examined the dependence of thermoregulation on colony size with temperature explicitly represented. Their Keller–Segel-type model couples local temperature T and bee density ρ :

$$\begin{cases} \frac{\partial T}{\partial t} = \Delta T + f(T)\rho, \\ \frac{\partial \rho}{\partial t} = \nabla[\nabla\rho - \chi(T)\rho\nabla T] - \theta(\rho, T)\rho. \end{cases} \quad (14)$$

Here, $f(T)$ describes the heat production by bees, $\chi(T)$ is a temperature-dependent thermotactic coefficient, and $\theta(\rho, T)$ represents individual mortality. The sign of $\chi(T)$ changes around the preferred temperature, allowing bees to move toward warmer regions when local temperatures are low and toward cooler regions when temperatures are high. The model identifies a critical colony size that separates two winter thermoregulatory states. Colonies above this threshold can maintain a warm core, whereas colonies below it remain cold. As mortality reduces the size of the colony during winter, crossing this threshold can produce rapid population loss. The critical population also increases as ambient temperature decreases, so colder conditions require larger colonies to maintain successful thermoregulation. Although Zeaiter and Myerscough [2020] and Bastiaansen et al. [2020] represent thermoregulation differently, both identify colony size as an important determinant of thermal stability and colony survival.

Temperature is only one component of environmental variation experienced by colonies. Seasonal changes also alter brood production, resource availability, and colony population dynamics. A nonautonomous honey bee model examined this temporal variation by introducing seasonality directly into the queen’s egg-laying rate [Chen et al., 2023]. The seasonal rate was written as

$$r(t) = r_0 \left[1 + \varepsilon \cos \left(\frac{2\pi(t - \psi)}{\gamma} \right) \right], \quad (15)$$

where r_0 is the average egg-laying rate, $\varepsilon \in (0, 1)$ measures the strength of seasonality, γ is the seasonal period, and ψ determines the timing of the maximum egg-laying rate. In the absence of mites, the model reduces to the bee-only system

$$\frac{du}{dt} = \bar{r}(t) \frac{u^2}{\widehat{K} + u^2} - \bar{d}_h u, \quad (16)$$

which combines seasonal reproduction with cooperative brood care and adult mortality. Analysis of this subsystem showed that seasonality does not have a uniformly positive or negative effect on colony persistence. Moderate seasonality can promote colony survival, whereas strong seasonality can increase the risk of collapse. The effect also depends on the timing of peak egg-laying, which can shift the boundary between colony survival and collapse.

This formulation represents seasonality through colony reproduction rather than through weather variables directly. Empirical studies provide a complementary connection between seasonal weather and colony survival. Switanek et al. [2017] analyzed winter mortality across Austria and found that warmer and drier conditions during the preceding year were associated with increased winter losses. They subsequently developed a statistical model using temperature and precipitation as predictors of colony mortality.

Calovi et al. [2021] used machine-learning models to predict overwintering survival from weather, topographic, landscape, and management variables in Pennsylvania. Their probability Random Forest model identified several weather variables, including growing degree days, maximum temperature of the warmest quarter, and precipitation, as important predictors of colony survival. A model based only on weather variables performed similarly to the model containing the full set of environmental and management predictors, highlighting the predictive importance of climatic conditions.

These data-driven approaches complement mechanistic population models: they can identify nonlinear relationships and environmental variables associated with colony survival without specifying the biological processes that generate those relationships.

[Rajagopalan et al., 2024] addressed this connection by coupling the VARROAPOP colony population model with downscaled climate projections. Their simulations showed that warmer autumn and

winter conditions can extend late-season flight rather than simply reduce cold stress. Continued flight physiologically ages workers and shifts the overwintering population toward older bees. When flight resumes in spring, losses of these older workers can occur before sufficient new brood has emerged, increasing the risk of spring colony failure. The model therefore links climate forcing to colony dynamics through changes in seasonal activity and worker age structure. Simulations also showed that indoor cold storage can reduce this effect by suppressing winter flight and limiting excessive worker aging.

The studies discussed above primarily emphasize winter conditions and seasonal weather. In comparison, short-term summer heat extremes and within-hive thermal variability are less commonly represented in mathematical population models. Chen et al. [2026] examined colony thermoregulation during a desert summer in which maximum shaded air temperatures intermittently exceeded 40°C. Average temperatures at the brood center and brood edge remained relatively stable and close to the optimal range of 34–36°C, but substantial diurnal fluctuations occurred within the hive. Brood also spent considerable portions of the day outside the optimal temperature range. Higher maximum air temperatures and greater within-hive temperature fluctuations were associated with declines in colony population. Larger colonies were also better able to buffer internal thermal variation.

These observations highlight a limitation of climate models that use only ambient temperature, flight suitability, or a single mean hive temperature. The spatial and temporal distribution of temperature within a colony may contain information that is lost in such summaries. This is also supported by Minaud et al. [2024], who monitored winter colony temperatures using multiple sensors and found that within-hive thermal amplitude distinguished surviving from dying colonies and provided an early indicator of colony collapse. A remaining modeling challenge is therefore to connect environmental temperature and weather variability with spatially heterogeneous hive temperatures and, in turn, with colony demographic processes. Such a model would link the external climate experienced by a colony to the internal thermal conditions actually experienced by brood and adult bees.

6.2 Agrochemical stress and exposure

Empirical work has established that agrochemicals reduce individual survival [Tsvetkov et al., 2017], impair learning and memory [Williamson and Wright, 2013], disrupt foraging and colony-level traits [Gill et al., 2012], reduce navigation and homing success [Henry et al., 2012], alter colony growth and reproductive output [Whitehorn et al., 2012], and interact with pathogens and parasites [Alaux et al., 2010, Doublet et al., 2015]. Landscape-scale studies extend these effects to wild bees and honey bees under realistic agricultural conditions [Rundlöf et al., 2015, Woodcock et al., 2017]. For social pollinators these individual impairments propagate through collective functions — brood care and nest thermoregulation [Crall et al., 2018], food collection and forager recruitment [Gill et al., 2012], worker productivity and colony demography [Bryden et al., 2013, Becher et al., 2014] — and ultimately to colony growth or failure. Metal contaminants provide an additional agrochemical exposure pathway in pollinators, including bumble bees [Gekièrè et al., 2024].

Agrochemical exposure can affect pollinators through both lethal and sublethal pathways, including changes in survival, foraging, navigation, reproduction, and colony growth. For social pollinators, these individual-level effects can propagate through colony demography and collective functions. Mathematical models have therefore focused increasingly on how pesticide effects at the individual or cohort level scale to colony-level population outcomes.

Early pesticide oriented models often represented agrochemical stress through changes in demographic rates or worker performance. [Bryden et al., 2013] modeled chronic sublethal stress as reduced worker performance and showed that persistent impairment of individuals can accumulate into colony failure. [Rumkee et al., 2015] used BEEHAVE to examine colony responses to increased mortality of larvae, in-hive workers, and foragers, as well as reduced egg laying. [Magal et al., 2019] considered pesticide-induced forager loss through increased homing failure, and later extended this framework to spatially distributed pesticide contamination in the foraging environment [Magal et al., 2020]. These models identified demographic and behavioral pathways through which pesticide exposure can contribute to

colony decline, although pesticide effects were represented primarily through imposed changes in mortality, reproduction, worker performance, or homing success.

Later colony models began to represent pesticide exposure more explicitly. Kuan et al. [2018] extended the VarroaPop framework with pesticide exposure and used Monte Carlo simulation and sensitivity analysis to examine foliar, soil, and seed-treatment scenarios. The model links pesticide concentrations in environmental and food compartments to contact and dietary exposure of different bee cohorts and translates exposure into mortality using stage-specific toxicity relationships. Their sensitivity analysis showed that queen strength and forager lifespan remained important determinants of colony dynamics, while pesticide-related parameters such as application rate, adult contact toxicity, and nectar load became important during exposure. The relative importance of these parameters also changed through time.

A related development occurred within BEEHAVE. Schmolke et al. [2019] added a pollen exposure effects module that connects clothianidin residues in treated cornfields with pollen collection, storage, consumption, and effects on larvae and adult bees. Because foraging occurs in an explicit landscape, colony exposure depends not only on residue concentration but also on the spatial and temporal availability of contaminated and uncontaminated resources. The simulations showed that landscape composition influenced the amount of contaminated corn pollen collected and the resulting colony-level effects, while effects on colony strength were observed primarily at residue concentrations above those measured in the field studies considered.

The VarroaPop pesticide framework was subsequently combined with statistical inference to estimate pesticide toxicity from colony-level data. Minucci et al. [2021] embedded VarroaPop+Pesticide within a Bayesian hierarchical framework and used Approximate Bayesian Computation with sequential Monte Carlo to estimate individual level toxicity parameters from a clothianidin colony feeding experiment. The fitted model captured the general population trends and declines observed at higher clothianidin concentrations, but underestimated effects at lower exposure and overestimated recovery at the highest exposure. These discrepancies indicated that oral toxicity induced mortality alone did not account for all observed colony level responses.

More recent models have expanded the representation of pesticide exposure and its movement through the colony. Garber et al. [2022] developed BeePop+ as an extension of the BeePop/VarroaPop framework for honey bee pesticide risk assessment. BeePop+ connects individual-level exposure and toxicity information with age-structured colony dynamics, weather, food resources, queen condition, and other stressors. This allows pesticide effects on different life stages to propagate to colony-level endpoints including brood, adult population, food stores, and colony survival.

Preuss et al. [2022] incorporated an ecotoxicological module into BEEHAVE to represent pesticide exposure and effects more explicitly. The resulting BEEHAVEecotox model includes external exposure, in-hive pesticide fate, water foraging, and effect modules. Pesticides can therefore be followed from contaminated nectar, pollen, water, or contact exposure into hive stores and subsequently to different adult and larval cohorts through food consumption. Dose response relationships then translate accumulated exposure into survival effects. This extends earlier BEEHAVE pesticide applications by resolving more of the pathway between environmental contamination and within-colony exposure rather than prescribing pesticide effects only through changes in demographic rates. Dawson et al. [2025] further refined this approach by incorporating a nectar based pesticide module into BEEHAVE. The module follows contaminated nectar collected by foragers into colony stores and subsequent exposure through nectar and honey consumption. These BEEHAVE based studies, together with BeePop+, illustrate a broader shift toward explicitly linking environmental pesticide residues, exposure of particular bee cohorts, dose response relationships, and colony level population dynamics.

A related but distinct direction integrates mathematical population models directly with experimental pesticide data. Chen et al. [2025] developed an age-structured delay differential equation model calibrated with experimental data on the fungicide Pristine[®]. Unlike exposure explicit simulation frameworks such as BeePop+ and BEEHAVEecotox, this approach does not attempt to resolve detailed pesticide

movement through environmental and within-hive compartments. Instead, it uses experimental treatment data to estimate how pesticide exposure alters colony demographic processes. The analysis showed that fungicide treatment affected both mortality and reproduction. Adult mortality increased with pesticide concentration, while the estimated timing of peak queen egg laying shifted under pesticide treatment. Thus, pesticide effects may alter not only survival but also the seasonal timing of colony recruitment. The model also estimated pollen consumption by larvae and adults, linking nutritional demand, pesticide exposure, and population dynamics. Simulations predicted colony decline and eventual collapse at sufficiently high pesticide concentrations, whereas control colonies maintained larger adult populations.

Although most mechanistic population models of agrochemical exposure have focused on honey bees, related ecotoxicological studies provide quantitative toxicity information for other pollinators. Gekière et al. [2024], for example, experimentally examined acute oral exposure to copper and cadmium in workers and males of the bumble bee *Bombus terrestris*. Dose–response and survival analyses showed that cadmium was substantially more toxic than copper. Apparent sex differences in cadmium LD₅₀ disappeared after accounting for bee mass, emphasizing the importance of body size when comparing toxicity among individuals. This study does not constitute a colony population model, but provides quantitative dose response information relevant to extending agrochemical modeling beyond honey bees.

A further extension considers pesticide exposure not only as an experimental treatment or prescribed environmental input, but as an endogenous component of an agroecosystem. Ongoing work by Rodriguez Rincon et al. couples honey bee colony size $H(t)$, plant population $P(t)$, pest population $V(t)$, and effective toxin load $T(t)$ in a deterministic ODE system. In contrast to models in which pesticide exposure is specified externally, pesticide input responds to pest abundance, creating feedback between pest pressure, pesticide accumulation, pollinator dynamics, and plant reproduction. The toxin dynamics can be written as

$$\frac{dT}{dt} = \underbrace{\kappa V}_{\text{pest-driven application}} - \underbrace{(u_V V + u_H H)T}_{\text{uptake}} - \underbrace{\delta_T T}_{\text{degradation}}. \quad (17)$$

Here, toxin input increases with pest abundance and is reduced through uptake and degradation. The coupled system examines conditions for mutualistic persistence, pest invasion, coexistence, and pesticide-induced collapse. Its bifurcation analysis also identifies bistability and Hopf bifurcations associated with pesticide feedback. Because this work is ongoing, these results should be interpreted as preliminary.

The models reviewed here differ substantially in scale and mathematical structure, but they also show a change in how agrochemical stress is represented. Early models imposed pesticide effects directly on worker performance or demographic rates. Later colony simulation models introduced explicit pathways from environmental residues to exposure and stage specific effects, while data calibrated models used observed colony responses to estimate pesticide dependent demographic or toxicity parameters. More recent work has begun to connect these processes to broader ecological interactions in which pesticide exposure itself may depend on pest and plant dynamics. An important next step is to combine these levels without losing biological interpretability, particularly by linking realistic exposure patterns with sublethal effects, colony demography, and interactions with other environmental stressors.

Several modeling gaps remain across climate and agrochemical stressors. For climate related stress, an important next step is to connect ambient temperature with the spatial and temporal variation of temperature within the hive and then determine how these thermal exposures affect colony demography. Existing models capture different components of this pathway, while field observations indicate that substantial within-hive temperature fluctuations can occur even when mean brood temperature remains near the optimal range. This suggests that future models may need to consider the magnitude, duration, and variability of thermal extremes rather than mean temperature alone. For agrochemical stress, existing models similarly capture different components of the pathway from environmental exposure to colony response. Further integration of realistic, time varying pesticide exposure with experimentally estimated effects on survival, reproduction, development, and behavior would help connect exposure dynamics more

directly with colony demography. Such models could also examine how pesticide exposure interacts with other stressors, including nutritional limitation and parasite pressure, in determining colony persistence. More broadly, integrating environmental exposure with internal colony conditions and demographic responses remains an important direction for connecting measurable environmental stressors with colony population dynamics.

7 Theme IV: Optimization of management interventions

The analytical structure developed above supports a reformulation of beekeeping management. Rather than treating intervention timing as a calendar rule, optimal control theory asks: given the current colony state — mite load, brood level, adult population, viral titre, season — what action maximizes a specified objective subject to the colony dynamics? The workshop’s central insight in this area was that *optimal management timing is a state-dependent decision, not a fixed date*.

7.1 Miticide timing as an optimal control problem

Let $u(t) \in [0, u_{\max}]$ denote acaricide application intensity. The controlled mite dynamics become

$$\frac{dM}{dt} = c\alpha HM - d_m M - u(t)M, \quad (18)$$

and the beekeeper’s problem is to minimize an integrated cost trading cumulative mite damage against treatment expense and the risk of queen or brood injury from aggressive treatment,

$$J(u) = \int_0^T [w_1 M(t) + w_2 H_{\text{loss}}(t) + w_3 u(t)^2] dt, \quad (19)$$

subject to the colony dynamics and non-negativity of all states. Applying Pontryagin’s maximum principle [Lenhart and Workman, 2007] yields adjoint equations whose costate variables characterize the optimal schedule $u^*(t)$ as a function of current colony state.

The structural finding is that the optimal schedule is *brood-phase dependent*: treatment is most effective, and carries the lowest injury risk, when the brood-to-adult ratio is low, because that is when the largest fraction of the mite population is phoretic and therefore exposed. This occurs naturally in late summer and early autumn as the colony begins pre-winter brood reduction. The practical significance is that the same quantity — brood-to-adult ratio — serves simultaneously as an early warning indicator (Section 9) and as the trigger variable for optimal treatment, giving it dual clinical utility. The current standard threshold of roughly one mite per three hundred bees is, on this analysis, a treatment *trigger* rather than an optimum, and optimization models can identify treatment windows as functions of colony trajectory rather than static counts.

7.2 Colony splitting and the viable-split problem

Colony splitting — dividing a strong colony into smaller units to prevent swarming and produce replacements — is a primary commercial management tool, and it poses a question that is mathematically transparent. For a split with initial brood b_0 and adults h_0 , the colony survives if and only if (b_0, h_0) lies in the basin of attraction of the viable equilibrium. Minimum viable split size is therefore not a biological constant but a *separatrix*, whose location depends on season, ambient mite load, nutritional environment, and queen status (mated, virgin, or absent).

This is one of the rare cases where a purely theoretical construct maps onto a directly executable field experiment. The workshop proposed creating splits of two, three, four, and five frames at multiple points in the season and monitoring survival biweekly, generating data from which the separatrix can be fitted. The working hypothesis is that two-frame splits generally fail, five-frame splits reliably succeed, and the three-to-four frame range is the decision zone — which is precisely the region in which the separatrix should be estimable with the greatest statistical power.

7.3 Nutritional supplementation

Extending the colony model with a nutritional state variable N gives

$$\frac{dN}{dt} = F(t) + S(t) - \frac{\gamma BN}{K_N + N} - \delta N, \quad (20)$$

where F is landscape-driven foraging intake (equation (??)), S is the supplemental feeding rate — the control variable — and γ, δ are consumption and degradation rates. Replacing the laying rate r in (2) by an increasing function $r(N)$ produces a *nutritionally modulated Allee effect* in which supplementation shifts the separatrix rather than simply adding biomass.

The relevance of this formulation has been sharpened by Moore et al. [2025], who showed that sterols absent from most commercial pollen substitutes are essential for sustained brood production, and that a diet incorporating an engineered yeast strain producing these sterols extended brood rearing substantially relative to sterol-deficient diets. This converts feeding protocol design into a genuine optimization problem with a compositional dimension: not merely when and how much to feed, but with what nutritional composition, subject to cost. It also raises a question the mathematics is well suited to answer — whether sterol supplementation shifts the separatrix enough to expand the viable split parameter space, and by how much.

7.4 The evaluation–optimization gap

Our inventory of post-2022 optimization literature (Appendix A, Table 7) supports a blunt assessment. Very few recent papers perform formal optimization. The overwhelming majority *evaluate* interventions — comparing treatment protocols, delivery matrices, or management regimes [Schödl et al., 2022, Panziera et al., 2023, Ssenoga et al., 2022] — rather than optimizing over a control space. *Varroa destructor* management dominates almost to the exclusion of other decisions. Very little work addresses colony splitting, queen replacement, supplemental feeding, overwintering strategy, or multi-objective management in which colony health and economic return are traded off explicitly.

This is a substantial and, importantly, an *accessible* gap. Optimal control for low-dimensional biological systems is mature methodology [Lenhart and Workman, 2007]; the barrier is not mathematical difficulty but the absence of well-posed objective functions grounded in beekeeping economics, and the absence of validated dynamics to optimize over. Both are addressable within the research agenda of Section 11.

A further constraint, articulated forcefully by commercial participants at the workshop, deserves to be stated as a modelling requirement rather than a practical afterthought. Commercial beekeepers need *decision rules*, not diagnostic dashboards. An optimal policy that requires solving a boundary value problem per colony will not be deployed. Deriving simplified, near-optimal feedback rules — and bounding their performance loss relative to the true optimum — is therefore a research objective in its own right, not a compromise.

Open Problem 7.1 (Multi-stressor optimal control). Formulate and solve the optimal control problem for simultaneous intervention on mites, nutrition, and thermal exposure, with controls representing acaricide application, supplemental feeding, and cold storage or shading. Determine whether the optimal policy decomposes approximately into independent single-stressor policies, or whether the interaction terms make joint optimization essential.

Open Problem 7.2 (Viable split separatrix). Derive analytical expressions for the basin boundary as a function of split size, queen status, ambient mite load, and season, and determine how nutritional supplementation — including sterol provision [Moore et al., 2025] — deforms it.

Open Problem 7.3 (Near-optimal feedback rules). For the control problem (18)–(19), construct simple threshold or piecewise-constant feedback policies expressible in quantities a beekeeper can measure, and derive bounds on their suboptimality relative to the Pontryagin solution.

Open Problem 7.4 (Economic portfolio optimization). Extend the control formulation to an operation managing many colonies under a shared labour and capital budget, where treating one colony forgoes treating another. Characterize the optimal allocation across a heterogeneous population of colonies, and determine when triage — abandoning the weakest colonies — is optimal.

8 Theme V: Spatial, landscape, and multispecies dynamics

The models considered so far treat the landscape as a fixed background — a parameter vector governing intake, nutritional quality, and exposure. In reality these inputs are spatially heterogeneous, temporally dynamic, and partly determined by the foraging behaviour of the colony itself. This theme is where the pollinator modelling literature is furthest from maturity, and where the extension beyond honey bees is most consequential.

8.1 Patch models and inter-colony dispersal

The simplest spatial extension couples colonies that exchange foragers and, critically, mites through forager drift. In the two-patch model of Messan et al. [2017],

$$\frac{dM_1}{dt} = c\alpha_1 H_1 M_1 - d_{m1} M_1 + \rho_{21} \left[\frac{H_2}{a_2 + H_2} \right] M_2 - \rho_{12} \left[\frac{H_1}{a_1 + H_1} \right] M_1, \quad (21)$$

where ρ_{ij} is the rate at which phoretic mites move from colony j to colony i . Three results carry direct management significance:

- *Intermediate* dispersal generates bistability between two interior equilibria with different mite loads in each colony — a mathematical explanation for the high between-colony variance in mite load routinely observed within a single apiary.
- *Large* dispersal destabilizes both colonies simultaneously, creating the possibility of apiary-level cascades in which one heavily infested collapsing colony rapidly exports mites to its neighbours.
- *One-directional* dispersal can stabilize the receiving colony at a higher mite load while destabilizing the source.

These results bear on apiary layout, management of forager drift, and the question of whether a collapsing colony should be removed or treated in place. None has been translated into practice, and a beekeeper-facing tool that accepts colony mite loads and apiary layout and returns a cascade risk assessment is an identified near-term deliverable.

8.2 Continuum models for foraging and pollination

At larger scales the distribution of foragers over a heterogeneous resource landscape is naturally described by reaction–diffusion or spatially structured integro-differential equations. Coupling the within-colony model to a spatial forager density $f(x, t)$ gives

$$\frac{\partial f}{\partial t} = D\nabla^2 f - \mu_f f + S(x, t)h(H, f), \quad (22)$$

where D is the forager diffusion coefficient, μ_f forager mortality, $S(x, t)$ the spatiotemporal floral resource landscape, and h a source term coupling forager density to within-colony adult population. Even existence and boundedness for the coupled ODE–PDE system are, to our knowledge, open. Magal et al. [2020] provide the most developed spatial treatment in the pesticide context, and Bastiaansen et al. [2020] the most developed within-hive spatial treatment, but the intermediate scale — colony coupled to landscape — remains largely unaddressed analytically.

8.3 Landscape data and machine learning for model inputs

Parameterizing $S(x, t)$ in (22) is a data problem before it is a mathematical one, and it is the problem the geospatial community has been solving independently. The Beescape and BeeSpatial platforms [Li et al., 2026, Robinson et al., 2021] deliver multi-scale environmental indicators of bee habitat — land cover, foraging and nesting resources, insecticide loading and toxicity, and climate trends — across the conterminous United States, with Beescape aimed at beekeepers, growers, and land managers, and BeeSpatial providing researchers with deeper data access and geoprocessing. These indicators build on the habitat-quality modelling framework of Lonsdorf et al. [2009] and have been used to relate landscape composition and climate to wild bee outcomes [Kammerer et al., 2021] and honey bee overwintering survival [Calovi et al., 2021].

Four additional data streams were identified at the workshop as tractable targets:

- deep learning applied to high-resolution aerial or drone imagery (0.1–1 m) to predict flowering species composition and abundance, with citizen science observations providing georeferenced training labels;
- phenological curve inference from time-series satellite reflectance (NDVI, EVI) to map seasonal bloom availability at 10–30 m resolution;
- automated waggle-dance decoding at scale, which provides direct, real-time evidence of where a colony is actually foraging and can validate landscape models rather than merely parameterize them [Jha and Kremen, 2013];
- cross-referencing of inferred foraging maps against pesticide application records to estimate realized field-level exposure.

Integrating these layers with colony dynamics requires inferring $S(x, t)$ from partially observed foraging behaviour — a high-dimensional Bayesian inverse problem suited to ensemble Kalman or variational data assimilation [Evensen, 2009].

8.4 Beyond *Apis mellifera*: multispecies and mutualism models

Two extensions take the framework beyond managed honey bees, and both matter for the pollinator decline problem as opposed to the colony collapse problem.

Other bee taxa. Bumble bees have a fundamentally different colony life history — annual cycles, solitary founding queens, much smaller colonies — which changes the mathematics. Bumble-BEEHAVE [Becher et al., 2018] provides a simulation framework at individual, colony, population, and community levels, and empirical work has established colony-level pesticide effects [Whitehorn et al., 2012, Crall et al., 2018] and landscape drivers of wild bee outcomes [Kammerer et al., 2021]. Analytically tractable models for annual social taxa, however, are largely absent. The Allee structure central to Section 4 takes a different form when the colony is founded by a single queen each year: the relevant threshold concerns queen production and overwintering survival, not the maintenance of a standing worker population.

Plant–pollinator mutualism. At community scale, the object of study is the mutualism itself. Consumer–resource formulations of density-dependent mutualism [Holland and DeAngelis, 2010] and models characterizing mutualists that compete at high density [Wang and Wu, 2011] provide the analytical vocabulary, while network approaches [Bascompte and Jordano, 2007] address the architecture of real communities and their robustness to phenological change and habitat loss [Revilla et al., 2015]. Obligate mutualisms generate Allee effects of their own: below a critical pollinator density, plant reproduction fails, which further reduces pollinator resources. The mathematical resonance with the within-colony Allee effect is not coincidental — both arise from cooperative, non-monotone per-capita response — and it suggests that the persistence and bifurcation machinery developed for colonies transfers to communities with modest

adaptation. The endogenous-pesticide model of Rodriguez Rincon et al. [2026], which couples pollinator, plant, pest, and toxin, is an example of exactly this transfer.

Open Problem 8.1 (Coupled ODE–PDE well-posedness). Establish existence, uniqueness, boundedness, and long-time behaviour for the coupled within-colony ODE and spatial forager PDE system (22), including the case of a moving colony boundary and heterogeneous, time-periodic resource landscape $S(x, t)$.

Open Problem 8.2 (Apiary network stability). Extend (21) to n colonies with heterogeneous dispersal kernel ρ_{ij} determined by apiary geometry. Determine the conditions on the network structure under which a single collapsing colony destabilizes the whole apiary, and identify layouts that minimize cascade risk subject to operational constraints.

Open Problem 8.3 (Managed–wild competition). Formulate a model in which managed honey bee colonies and wild bee populations compete for a shared, spatially explicit floral resource, and determine conditions under which managed colony stocking density reduces wild bee persistence. Given that sterol-supplemented diets may reduce honey bee reliance on natural pollen [Moore et al., 2025], quantify the extent to which supplementary feeding could relax this competition.

Open Problem 8.4 (Mutualism collapse thresholds). For plant–pollinator systems with obligate dependence on at least one side, characterize the conditions for unique, globally attracting coexistence and the thresholds at which the mutualism collapses. Determine how these thresholds respond to phenological drift and to pollinator population decline driven by the within-colony mechanisms of Sections 4–6.

9 Cross-cutting methodology: data, early warning, and hybrid modelling

9.1 Critical slowing down as a theoretical foundation for early warning

A general result of dynamical systems theory is that systems approaching a bifurcation exhibit characteristic changes in their statistical properties, collectively termed *critical slowing down* [Scheffer et al., 2012, Dakos et al., 2012]. For a scalar system $\frac{dH}{dt} = f(H, \mu)$ near a saddle-node bifurcation at $\mu = \mu_c$, the linearization about the equilibrium H^* gives

$$\frac{d(\delta H)}{dt} \approx \lambda(\mu) \delta H, \quad \lambda(\mu) = \left. \frac{\partial f}{\partial H} \right|_{H^*} \rightarrow 0 \quad \text{as } \mu \rightarrow \mu_c. \quad (23)$$

As the dominant eigenvalue approaches zero, perturbations decay more slowly. Under the stochastic forcing of (??), this implies that both the variance and the lag-1 autocorrelation of $H(t)$ increase as the bifurcation is approached.

For colonies this is a substantive and testable prediction: rising autocorrelation in acoustic activity or hive temperature, or rising variance in brood-to-adult ratio, should precede observable demographic decline. Because the mechanisms of Section 4 place saddle-node and subcritical Hopf bifurcations at the heart of collapse, early warning theory is not an add-on to the colony modelling programme — it is a direct corollary of it.

Two qualifications matter. First, critical slowing down is a property of the approach to a bifurcation, not of stochastic crossing of a fixed threshold; a colony driven across the separatrix by a single extreme event may exhibit no precursor at all. Second, the lead time achievable depends on the rate of parameter drift relative to noise intensity, and this trade-off has not been quantified for any bee system.

9.2 Sensor data and the empirical detection pipeline

High-frequency in-hive monitoring makes empirical testing of this theory feasible for the first time. The dataset presented at the workshop by Fine (USDA ARS) — one year of 15-minute-resolution

internal acoustic, temperature, and humidity measurements from 40 colonies, paired with monthly manual population and brood counts, with a subset of colonies experiencing collapse — is unusual in combining temporal resolution, biological ground truth, and observed failure events.

The analytical pipeline proposed at the workshop has four stages:

1. **Preprocessing.** Seasonal-trend decomposition (e.g. STL) to separate the residual stochastic component from daily, weekly, and seasonal periodicities and to handle sensor artefacts.
2. **Indicator computation.** Rolling-window estimates of lag-1 autocorrelation, variance, and spectral reddening for each stream, combined into a multi-stream indicator score.
3. **Lead-time analysis.** For colonies that failed, the point at which the indicator first exceeded the control distribution; lead time is the primary performance metric.
4. **Threshold optimization.** ROC analysis to select the operating point, weighted by the *economic* costs of false alarms and missed detections rather than by statistical convention alone.

Brood-to-adult and mite-to-adult ratios are expected to be particularly informative, both because they measure proximity to the critical transitions directly and because, as noted in Section 7, low brood-to-adult periods coincide with the optimal acaricide window.

9.3 Detection and monitoring: a rapidly growing empirical literature

Independently of dynamical early warning theory, a substantial applied literature has developed around automated detection, inventoried in Appendix A, Table 8. It divides into several strands: non-invasive *Varroa destructor* detection from brood-comb vibration [Hall et al., 2023], from images via deep learning [Divasón et al., 2023], and from hyperspectral imagery [Duma et al., 2024]; integrated hive-entrance monitoring of bees, pollen, and mites [Narcia-Macias et al., 2023]; acoustic detection of queenlessness [Ruvinga et al., 2023, Kanelis et al., 2023]; hive weight and environmental modelling of colony condition [Degenfellner and Templ, 2024]; and temperature-based prediction of overwintering mortality [Minaud et al., 2024]. Acoustic monitoring for exposure assessment has a longer history [Bromenshenk et al., 2009].

An honest assessment of this literature is that most of it performs *detection* of a current state rather than *prediction* of a future one. Detecting that a colony is already queenless, or already infested, is operationally valuable but is not early warning in the dynamical sense. The papers that come closest to genuine prediction — overwintering mortality from temperature series, colony condition from weight trajectories — are precisely those that use time series structure rather than instantaneous classification. This suggests where effort should go.

9.4 Hybrid mechanistic–machine learning approaches

The central methodological tension in this field is between the richness of the theoretical framework and the sparsity of the data. Fitting a seven-stressor, multi-phase system to monthly measurements from a few dozen colonies is not a well-posed estimation problem. Hybrid approaches, which retain mechanistic structure while learning unknown functional relationships, are the emerging response.

Neural differential equations. A neural ODE [Chen et al., 2018] replaces specified functional forms with learned representations,

$$\frac{d}{dt} \begin{pmatrix} H \\ B \\ M \\ V \end{pmatrix} = f_{\theta}(H, B, M, V, t), \quad (24)$$

where f_{θ} is trained on observed trajectories. The value of this for biology lies in constrained hybridization: f_{θ} can be structured to enforce non-negativity, to reproduce the Allee form of (2), and to respect the mite

phase transition, while the *synergy* function coupling stressors — which is exactly what theory does not specify — is learned from data. Physics-informed approaches [Raissi et al., 2019] offer a complementary route in which the mechanistic model enters as a soft constraint in the loss.

Data assimilation. Colonies are partially observed: adult population by frame count (coarse), mite load by alcohol wash (destructive, infrequent), brood by visual inspection, viral load by PCR (expensive). Sensors supply high-frequency signals with no direct mapping to state variables. Data assimilation maintains the conditional distribution $p(x_t | y_{1:t})$, where $x_t = (H_t, B_t, M_t, V_t, N_t, \dots)$ is the colony state and $y_{1:t}$ the observation history, with the transition density supplied by the mechanistic model and the observation model mapping states to measurable quantities [Evensen, 2009]. This yields real-time state estimation from heterogeneous streams, calibrated uncertainty about unobserved variables, and intervention timing based on a full posterior rather than a noisy point estimate.

Latent states. Queen health, cumulative pesticide exposure, and genetic stress resistance affect dynamics but are not measured. Latent variable models — hidden Markov models, latent Gaussian processes, variational autoencoders — can in principle infer them from observable consequences. Whether they *can* in practice is an identifiability question, and it is open.

Open Problem 9.1 (Early warning lead-time bounds). Derive bounds on the maximum lead time achievable by any indicator for the saddle-node and subcritical Hopf bifurcations of Section 4, as a function of noise intensity σ and the rate of parameter drift toward the bifurcation. Determine whether fusing acoustic, thermal, and manual monitoring streams achieves provably better lead times than any single stream, and characterize the regimes in which no useful warning is possible.

Open Problem 9.2 (Identifiability of hybrid models). Establish conditions under which the latent parameters of a hybrid mechanistic–latent-state colony model are consistently estimable from the data streams available in standard monitoring protocols. Determine the minimum observation frequency and diversity required for identifiability, and identify which latent variables are provably non-identifiable.

10 Synthesis: structural gaps and the open-problem agenda

10.1 Four structural limitations of the current literature

Reading across the five themes, four limitations recur with enough consistency to be regarded as structural rather than incidental.

G1. Multi-stressor integration. Existing models incorporate at most two stressor categories simultaneously. The seven-stressor system of Table 1 has never been analysed as an integrated dynamical system, and the few genuine multi-stressor efforts [Osorio Rodríguez and Toro Zapata, 2025] pair a pathogen with a single environmental factor. This matters because the empirical case for interaction is strong, and because the mathematics of Sections 4–5 shows that stressor effects combine within scalar bifurcation criteria rather than additively.

G2. Data–model coupling. Rigorous fitting and validation have begun in stage-structured and pesticide-exposure contexts [Chen et al., 2020, 2025], but comprehensive datasets tracking mite load, viral titre, nutritional status, demographic structure, and high-frequency sensor streams *simultaneously* remain rare. No single existing dataset captures all seven stressors across multiple colonies over a full annual cycle. This is the field’s most binding near-term constraint, and notably it is an infrastructure problem rather than a scientific one.

G3. Scale disconnection. Within-hive spatial models, colony demographic models, and landscape models each exist and are individually sophisticated, but translation between them is largely absent. The colony appears in landscape models as a point sink and in within-hive models as a boundary condition. The intermediate coupling — colony dynamics driven by a resolved landscape and a resolved internal environment — is where the predictive gains lie and where the analysis is hardest.

G4. Translation to practice. The gap between mathematical results and operational tools is wide. Closing it requires engaging the economic, operational, and scale constraints of commercial beekeeping as *modelling constraints* from the outset. A policy that cannot be executed by a crew managing thousands of colonies is not, for this purpose, a solution.

To these we would add a fifth, which is about people rather than mathematics: few researchers combine the analytical training this problem requires with the entomological knowledge needed to formulate biologically meaningful models. The workshop network is itself a partial remedy.

10.2 Consolidated open problems

Table 3 collects the sixteen open problems posed above, ordered by approximate technical accessibility — from problems addressable with existing tools to those requiring substantial new development. We regard P1, P4, P8, and P11 as the highest-value entry points: each is tractable with current methods, each has an identified dataset against which it could be validated, and each unlocks several others.

Table 3: Consolidated open mathematical problems, with domain and significance. Numbering follows order of appearance.

#	Problem	Domain	Significance
P1	Basin boundary characterization (OP 4.1)	Bifurcation theory, dynamical systems	Foundation for composite stress score and state-dependent management
P2	Mechanism identification (OP 4.2)	Inverse problems, identifiability	Determines whether monitoring can distinguish collapse pathways
P3	Multi-virus coinfection (OP 5.1)	Multi-strain epidemic theory	Most colonies carry several viruses; single-virus models miss interactions
P4	Cross-taxon parasite interaction (OP 5.2)	Coupled epidemic systems	<i>Varroa destructor</i> – <i>Nosema</i> interaction is unmodelled despite joint surveillance
P5	Treatment-induced destabilization (OP 5.3)	Perturbation of attractors	Determines whether standard acaricide protocols can precipitate collapse
P6	Stochastic collapse-time distribution (OP ??)	SDEs, first-passage theory	Quantifies how climate variability accelerates collapse risk
P7	Pesticide catastrophe criterion (OP ??)	Bifurcation theory, toxicology	Would give risk assessment a dynamical foundation
P8	Critical phenological offset (OP ??)	Nonautonomous dynamics	Quantifies climate risk of nutritional decoupling
P9	Thermal exposure functionals (OP ??)	Time series, model reduction	Mean temperature demonstrably underestimates heat stress
P10	Multi-stressor optimal control (OP 7.1)	Optimal control, Pontryagin	Foundation for joint intervention protocols
P11	Viable split separatrix (OP 7.2)	Basin geometry	Directly testable by field experiment; immediate commercial value
P12	Near-optimal feedback rules (OP 7.3)	Control approximation	Required for any deployable decision tool
P13	Economic portfolio optimization (OP 7.4)	Stochastic control, operations research	Reflects how commercial operations actually allocate effort
P14	Coupled ODE–PDE well-posedness (OP 8.1)	PDEs, functional analysis	Prerequisite for rigorous colony–landscape models
P15	Apiary network stability (OP 8.2)	Network dynamical systems	Determines when one collapsing colony threatens an apiary
P16	Managed–wild competition and mutualism collapse (OP 8.3, 8.4)	Mutualism theory, monotone systems	Extends the agenda from colony collapse to pollinator decline

Also open: early warning lead-time bounds (OP 9.1); hybrid model identifiability (OP 9.2).

11 A research agenda

The programme sketched here requires coordination that did not previously exist. We set it out in three phases, with the caveat that phase boundaries are logical rather than strictly temporal and that delays in the first phase propagate.

11.1 Phase I: shared infrastructure

The most urgent need is not new data but access to existing data. Multiple high-quality datasets exist across participating laboratories and have not been integrated. Priority actions are:

- a shared repository and public-facing project site providing access to data, model code, and tools, with links to existing resources including Beescape and BeeSpatial [Li et al., 2026] and the BeePop+/VarroaPop lineage [Garber et al., 2022];
- release of multi-year colony measurement data (frame counts, *Varroa destructor* levels, pollen and virus data) organized against the corresponding publications;
- a feasibility assessment and data-use protocol for sharing high-frequency in-hive sensor datasets;
- a *dataset crosswalk* documenting, for every available dataset, the variables measured, temporal resolution, colony count, geographic coverage, and format — this is the single deliverable that most directly unblocks the rest;
- a power analysis determining sample sizes and measurement frequency for a future large-scale monitoring study;
- sub-teams around the priority areas, with leads and milestones.

11.2 Phase II: pilot studies

Six pilot tracks correspond to the priority areas and are designed to produce early, visible outputs and the preliminary results that competitive proposals require (Table 4).

Table 4: Pilot study tracks, linked to the open problems they address.

Track	Objective	Problems
Baseline colony model	Fit the Khoury–Perry ODE framework with modular <i>Varroa destructor</i> and nutrition extensions to multi-site colony data; sensitivity analysis identifying the parameters most predictive of survival versus collapse	P1, P2
Resilience and tipping points	Estimate the composite stress score Σ for each colony using the bifurcation criterion; validate against known outcomes; compare predictive power against current monitoring thresholds	P1, P6
Early warning	Apply seasonal-trend decomposition and rolling indicator analysis to high-frequency sensor data; compute lead-time distributions for acoustic and thermal indicators; validate against observed collapse	P9, and OP 9.1
Intervention optimization	Solve the Pontryagin problem for acaricide timing with parameters fit to colony data; compare the optimal schedule against calendar protocols; run the split-size field experiment	P10–P12
Climate and landscape	Fit the seasonal model (??) to multi-year data; estimate the change in collapse probability per degree of summer warming; couple to landscape indicators	P6, P8, P9
Hybrid machine learning	Train a constrained neural ODE on a held-out split of the sensor dataset; compare predictive accuracy against the mechanistic model alone and against a purely data-driven baseline	OP 9.2

Beyond these, two synthesis products are worth naming: a *baseline colony model technical report* establishing a shared mathematical reference with modular stressor extensions, so that sub-teams build

on a common foundation rather than incompatible variants; and this review itself, as a statement of the agenda.

11.3 Phase III: integrated programme development

Pilot results support coordinated proposals across several mechanisms: mathematical biology programmes for multi-stressor resilience modelling; agricultural research programmes for a large-scale monitoring study capturing all seven stressor categories across multiple sites and years; integrative organismal–mathematical mechanisms for colony–landscape coupling; and translational or convergence mechanisms for decision-support tools. Formal data-sharing and co-development agreements with commercial operations belong in this phase as well, and are as much a scientific requirement — field validation — as a funding strategy.

11.4 Longer-term objectives

Over a three-year horizon the programme should deliver: a validated, modular, open-source framework for colony dynamics incorporating all seven stressor categories; a near-real-time monitoring and risk assessment capability combining sensor data with model prediction and delivering actionable guidance; extension of the tools to bumble bees, solitary bees, and native bee communities; and a sustained training pipeline for researchers at the intersection of mathematical biology, entomology, and data science.

12 Concluding remarks

Colony losses have remained at historically anomalous levels for two decades despite sustained research and intensive management. That persistence is itself evidence that the problem has not been characterized at the level at which it operates. Our central claim is that it operates at the level of dynamical systems: interacting stressors generating threshold effects, bifurcations, and catastrophic transitions in a state space that cannot be understood through linear, single-stressor, or purely descriptive approaches.

Five conclusions follow from the material reviewed here.

1. **Collapse is an attractor transition.** Colony failure is most rigorously understood as a stochastically driven transition from the viable to the extinction attractor. The distance from the colony’s state to the basin boundary — not the level of any single stressor — governs vulnerability. This reframing has an immediate practical consequence: monitoring should estimate proximity to a boundary, not compare individual measurements to individual thresholds.
2. **Synergy is the central problem, and it is structural.** Multi-stressor models are not merely more complete versions of single-stressor models. They exhibit qualitatively different behaviour — multiple attractors, disease persistence at sub-threshold mite loads, parasite-mediated stabilization. The mathematics shows why: stressors enter scalar bifurcation criteria such as (7) multiplicatively.
3. **The same structures recur across scales and taxa.** Non-monotone cooperative response generating Allee effects appears in colony demography, in thermoregulation, in winter cluster spatial organization, and in plant–pollinator mutualism. This is an opportunity: results and techniques transfer, and the pollinator decline problem is more unified mathematically than its fragmented literature suggests.
4. **The binding constraint is data integration, not data collection.** High-quality datasets exist. What is missing is the infrastructure and the analytical frameworks to combine them. Hybrid mechanistic–machine-learning methods can extract substantial value from existing heterogeneous, incomplete data without waiting a decade for an ideal prospective study.
5. **Translation must be designed in.** Models will improve bee health only if they yield tools that beekeepers, land managers, and policymakers can use. Simplification for deployment, and bounding

the cost of that simplification, is a research objective rather than a concession.

We close with the methodological principle that emerged most consistently from the workshop: *staged model complexity*. Simple ODE models of bee–mite interaction reproduce the qualitative dynamics of far more complex delay and age-structured systems. This licenses a strategy of beginning with analytically tractable models, extracting structural results — bifurcation criteria, stability conditions, the form of optimal controls — and adding mechanism modularly only where the simpler model demonstrably fails to account for observation. The discipline matters because it keeps mathematical complexity in the service of biological insight rather than in competition with it.

Honey bees and their wild relatives have persisted for tens of millions of years. The crisis they now face is a product of the last eighty, and it is therefore a problem that belongs to our era and to our responsibility. The mathematical tools required to understand it with the necessary rigour largely exist. They have simply not yet been brought to bear systematically. That is the gap this review, and the programme it describes, are intended to close.

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References

- Samuel Abban, Bart Smith, Miguel Corona, Steven C. Cook, Jay D. Evans, Yanping Chen, and Mohamed Alburaki. Prevalence and distribution of *Varroa destructor* and *Nosema* spp. in symptomatic honey bee colonies across the USA from 2015 to 2022. *Scientific Reports*, 14:1726, 2024. doi: 10.1038/s41598-024-51514-9.
- AIM Workshop Participants. Addressing declining pollinator populations through new mathematics: Workshop final report. Technical report, American Institute of Mathematics, Pasadena, CA, 2026. Workshop held March 30–April 5, 2026. <https://aimath.org/pastworkshops/year2026>.
- Cédric Alaux, Jean-Luc Brunet, Claudia Dussaubat, Fanny Mondet, Sylvie Tchamitchan, Marianne Cousin, Julien Brillard, Aurélie Baldy, Luc P. Belzunces, and Yves Le Conte. Interactions between *Nosema* microspores and a neonicotinoid weaken honeybees (*Apis mellifera*). *Environmental Microbiology*, 12(3):774–782, 2010. doi: 10.1111/j.1462-2920.2009.02123.x.
- Mohammed H. Alharbi. Mathematical analysis for honeybee dynamics under the influence of seasonal environments. *Mathematics*, 12(22):3496, 2024. doi: 10.3390/math12223496.
- Warder C. Allee. *Animal Aggregations: A Study in General Sociology*. University of Chicago Press, Chicago, 1931.
- Gro V. Amdam, Kari Norberg, Robert E. Page, M. Kim Fondrk, and Joachim Erber. Downregulation of vitellogenin gene activity increases the gustatory responsiveness of honey bee workers. *Behavioural Brain Research*, 169(2):201–205, 2006. doi: 10.1016/j.bbr.2006.01.006.
- [Authors to verify]. Threshold dynamics of honey bee colonies with general eclosion and recruitment rates. *Axioms*, 14(10):773, 2025. Authorship requires verification before submission.
- Jordi Bascompte and Pedro Jordano. Plant–animal mutualistic networks: the architecture of biodiversity. *Annual Review of Ecology, Evolution, and Systematics*, 38:567–593, 2007. doi: 10.1146/annurev.ecolsys.38.091206.095818.
- Robbin Bastiaansen, Arjen Doelman, Frank van Langevelde, and Vivi Rottschäfer. Modeling honey bee colonies in winter using a Keller–Segel model with a sign-changing chemotactic coefficient. *SIAM Journal on Applied Mathematics*, 80(2):839–863, 2020. doi: 10.1137/19M1246067.
- Matthias A. Becher, Volker Grimm, Pernille Thorbek, Juliane Horn, Peter J. Kennedy, and Juliet L. Osborne. BEEHAVE: a systems model of honeybee colony dynamics and foraging to explore multifactorial causes of colony failure. *Journal of Applied Ecology*, 51(2):470–482, 2014. doi: 10.1111/1365-2664.12222.
- Matthias A. Becher, Grace Twiston-Davies, Tim D. Penny, Dave Goulson, Ellen L. Rotheray, and Juliet L. Osborne. Bumble-BEEHAVE: a systems model for exploring multifactorial causes of bumblebee decline at individual, colony, population and community level. *Journal of Applied Ecology*, 55(6): 2790–2801, 2018. doi: 10.1111/1365-2664.13165.
- Sara Bernardi and Ezio Venturino. Viral epidemiology of the adult *Apis mellifera* infested by the *Varroa destructor* mite. *Heliyon*, 2(5):e00101, 2016. doi: 10.1016/j.heliyon.2016.e00101.
- Matthew I. Betti, Lindi M. Wahl, and Mair Zamir. Effects of infection on honey bee population dynamics: a model. *PLOS ONE*, 9(10):e110237, 2014. doi: 10.1371/journal.pone.0110237.
- Tjeerd Blacquière, Guy Smagghe, Cornelis A. M. van Gestel, and Veerle Mommaerts. Neonicotinoids in bees: a review on concentrations, side-effects and risk assessment. *Ecotoxicology*, 21(4):973–992, 2012. doi: 10.1007/s10646-012-0863-x.

- Jerry J. Bromenshenk, Colin B. Henderson, Robert A. Seccomb, Steven D. Rice, Robert T. Etter, Susan F. A. Bender, Philip J. Rodacy, Joseph A. Shaw, Nathan L. Seldomridge, Lee H. Spangler, and Jay J. Wilson. Honey bee acoustic monitoring to assess the risk or actuality of bee poisoning. *Apidologie*, 40(4):445–462, 2009. Title and pagination require verification.
- Selina Bruckner, Mikayla Wilson, Dan Aurell, Karen Rennich, Dennis vanEngelsdorp, Nathalie Steinhauer, and Geoffrey R. Williams. A national survey of managed honey bee colony losses in the USA: results from the Bee Informed Partnership for 2017–18, 2018–19, and 2019–20. *Journal of Apicultural Research*, 62(3):429–443, 2023. doi: 10.1080/00218839.2022.2158586.
- Laura M. Brutscher, Katie F. Daughenbaugh, and Michelle L. Flenniken. Antiviral defense mechanisms in honey bees. *Current Opinion in Insect Science*, 10:71–82, 2015. doi: 10.1016/j.cois.2015.04.016.
- John Bryden, Richard J. Gill, Robert A. A. Mitton, Nigel E. Raine, and Vincent A. A. Jansen. Chronic sublethal stress causes bee colony failure. *Ecology Letters*, 16(12):1463–1469, 2013. doi: 10.1111/ele.12188.
- Martina Calovi, Christina M. Grozinger, Douglas A. Miller, and Sarah C. Goslee. Summer weather conditions influence winter survival of honey bees (*Apis mellifera*) in the northeastern United States. *Scientific Reports*, 11:1553, 2021. doi: 10.1038/s41598-021-81051-8.
- Jun Chen, Komi Messan, Marisabel Rodriguez Messan, Gloria DeGrandi-Hoffman, Dingyong Bai, and Yun Kang. How to model honeybee population dynamics: stage structure and seasonality. *Mathematics in Applied Sciences and Engineering*, 1(2):91–125, 2020. doi: 10.5206/mase/10559.
- Jun Chen, Gloria DeGrandi-Hoffman, Vardayani Ratti, and Yun Kang. Review on mathematical modeling of honeybee population dynamics. *Mathematical Biosciences and Engineering*, 18(6):9606–9650, 2021. doi: 10.3934/mbe.2021471.
- Jun Chen, Jhonatan O. Rodriguez Rincon, Gloria DeGrandi-Hoffman, Jennifer Fewell, Jon Harrison, and Yun Kang. Impacts of seasonality and parasitism on honey bee population dynamics. *Journal of Mathematical Biology*, 87(1):19, 2023. doi: 10.1007/s00285-023-01952-2.
- Jun Chen, Adrian Fisher II, Jon F. Harrison, Gloria DeGrandi-Hoffman, Jennifer H. Fewell, and Yun Kang. Modeling pesticide impacts on honey bee dynamics from mathematical and experimental integration. *Journal of Theoretical Biology*, 611:112167, 2025. doi: 10.1016/j.jtbi.2025.112167.
- Jun Chen, Adrian Fisher II, Gloria DeGrandi-Hoffman, Cahit Ozturk, Brian H. Smith, Jennifer H. Fewell, Yun Kang, K. Maxwell, K. Overcash, K. Chahal, and Jon F. Harrison. Negative effects of excessive heat on colony thermoregulation and population dynamics in honeybees. *Ecological and Evolutionary Physiology*, 99(1), 2026. doi: 10.1086/739493.
- Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. *Advances in Neural Information Processing Systems*, 31:6571–6583, 2018.
- James D. Crall, Callin M. Switzer, Robert L. Oppenheimer, Ashlee N. Ford Versypt, Biswadip Dey, Andrea Brown, Mackay Eyster, Claire Guérin, Naomi E. Pierce, Stacey A. Combes, and Benjamin L. de Bivort. Neonicotinoid exposure disrupts bumblebee nest behavior, social networks, and thermoregulation. *Science*, 362(6415):683–686, 2018. doi: 10.1126/science.aat1598.
- Benjamin Dainat, Jay D. Evans, Yan Ping Chen, Laurent Gauthier, and Peter Neumann. Predictive markers of honey bee colony collapse. *PLOS ONE*, 7(2):e32151, 2012. doi: 10.1371/journal.pone.0032151.
- Vasilis Dakos, Stephen R. Carpenter, William A. Brock, Aaron M. Ellison, Vishwesha Guttal, Anthony R. Ives, Sonia Kéfi, Valerie Livina, David A. Seekell, Egbert H. van Nes, and Marten Scheffer. Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLOS ONE*, 7(7):e41010, 2012. doi: 10.1371/journal.pone.0041010.

- Daniel E. Dawson, Jeffrey M. Minucci, and S. Thomas Purucker. Incorporating a nectar-based pesticide module into BeeHave, an agent-based model of honeybee population dynamics. *Ecological Solutions and Evidence*, 6:e70105, 2025. doi: 10.1002/2688-8319.70105.
- Jurgen Degenfellner and Matthias Templ. Modeling bee hive dynamics: assessing colony health using hive weight and environmental parameters. *Computers and Electronics in Agriculture*, 218:108742, 2024. doi: 10.1016/j.compag.2024.108742.
- Gloria DeGrandi-Hoffman, Scott A. Roth, G. L. Loper, and E. H. Erickson. BEEPOP: a honeybee population dynamics simulation model. *Ecological Modelling*, 45(2):133–150, 1989. doi: 10.1016/0304-3800(89)90088-4.
- Garance Di Pasquale, Marion Salignon, Yves Le Conte, Luc P. Belzunces, Axel Decourtye, André Kretzschmar, Séverine Suchail, Jean-Luc Brunet, and Cédric Alaux. Influence of pollen nutrition on honey bee health: do pollen quality and diversity matter? *PLOS ONE*, 8(8):e72016, 2013. doi: 10.1371/journal.pone.0072016.
- Jose Divasón, Francisco J. Martínez-de Pisón, Ana Romero, Pilar Santolaria, and Jesús L. Yániz. Varroa mite detection using deep learning techniques. In *Hybrid Artificial Intelligent Systems*, volume 14001 of *Lecture Notes in Computer Science*, pages 326–337. Springer, 2023. doi: 10.1007/978-3-031-40725-3_28.
- Philip Donkersley, Glenn Rhodes, Roger W. Pickup, Kevin C. Jones, and Kenneth Wilson. Honeybee nutrition is linked to landscape composition. *Ecology and Evolution*, 4(21):4195–4206, 2014. doi: 10.1002/ece3.1293.
- Vincent Doublet, Maureen Labarussias, Joachim R. de Miranda, Robin F. A. Moritz, and Robert J. Paxton. Bees under stress: sublethal doses of a neonicotinoid pesticide and pathogens interact to elevate honey bee mortality across the life cycle. *Environmental Microbiology*, 17(4):969–983, 2015. doi: 10.1111/1462-2920.12426.
- Zibusiso-Sky Duma, Tomas Zemcik, Simon Bilik, Tuomas Sihvonen, Peter Honec, Satu-Pia Reinikainen, and Karel Horak. Varroa destructor detection on honey bees using hyperspectral imagery. *Computers and Electronics in Agriculture*, 224:109219, 2024. doi: 10.1016/j.compag.2024.109219. Volume and article number require verification.
- Hermann J. Eberl, Mallory R. Frederick, and Peter G. Kevan. Importance of brood maintenance terms in simple models of the honeybee–Varroa destructor–acute bee paralysis virus complex. *Electronic Journal of Differential Equations*, Conference 19:85–98, 2010.
- Miled El Hajji et al. Modeling dual-colony Nosema transmission in honeybees: the role of distributed delays and antiviral treatment. *AIMS Mathematics*, 11(1):2645–2681, 2026. doi: 10.3934/math.2026107. Full author list requires verification.
- Geir Evensen. *Data Assimilation: The Ensemble Kalman Filter*. Springer-Verlag, Berlin, 2nd edition, 2009.
- Kristina Garber, Gloria DeGrandi-Hoffman, Robert Curry, Jeffrey M. Minucci, Daniel E. Dawson, Cameron Douglass, Joseph P. Milone, and S. Thomas Purucker. Simulating the effects of pesticides on honey bee (*Apis mellifera* L.) colonies with BeePop+. *Ecologies*, 3(3):275–291, 2022. doi: 10.3390/ecologies3030022.
- Antoine Gekière, Luna Breuer, Loïc Dorio, Maryse Vanderplanck, and Denis Michez. Lethal effects and sex-specific tolerance of copper and cadmium in the buff-tailed bumble bee. *Environmental Toxicology and Pharmacology*, 110:104546, 2024. doi: 10.1016/j.etap.2024.104546.

- Richard J. Gill, Oscar Ramos-Rodriguez, and Nigel E. Raine. Combined pesticide exposure severely affects individual- and colony-level traits in bees. *Nature*, 491(7422):105–108, 2012. doi: 10.1038/nature11585.
- Dave Goulson, Elizabeth Nicholls, Cristina Botías, and Ellen L. Rotheray. Bee declines driven by combined stress from parasites, pesticides, and lack of flowers. *Science*, 347(6229):1255957, 2015. doi: 10.1126/science.1255957.
- Helen Hall, Martin Bencsik, and Michael Newton. Automated, non-invasive Varroa mite detection by vibrational measurements of gait combined with machine learning. *Scientific Reports*, 13:10202, 2023. doi: 10.1038/s41598-023-36810-0.
- Mickaël Henry, Maxime Béguin, Fabrice Requier, Orianne Rollin, Jean-François Odoux, Pierrick Aupinel, Jean Aptel, Sylvie Tchamitchian, and Axel Decourtye. A common pesticide decreases foraging success and survival in honey bees. *Science*, 336(6079):348–350, 2012. doi: 10.1126/science.1215039.
- J. Nathaniel Holland and Donald L. DeAngelis. A consumer–resource approach to the density-dependent population dynamics of mutualism. *Ecology*, 91(5):1286–1295, 2010. doi: 10.1890/09-1163.1.
- Mahmoud A. Ibrahim and Attila Dénes. A mathematical model for the spread of Varroa mites in honeybee populations: two simulation scenarios with seasonality. *Heliyon*, 8(10):e10648, 2022. doi: 10.1016/j.heliyon.2022.e10648.
- Shalene Jha and Claire Kremen. Resource diversity and landscape-level homogeneity drive native bee foraging. *Proceedings of the National Academy of Sciences*, 110(2):555–558, 2013. doi: 10.1073/pnas.1208682110.
- Melanie Kammerer, Sarah C. Goslee, Margaret R. Douglas, John F. Tooker, and Christina M. Grozinger. Wild bees as winners and losers: relative impacts of landscape composition, quality, and climate. *Global Change Biology*, 27(6):1250–1265, 2021. doi: 10.1111/gcb.15485.
- Dimitrios Kanelis, Vasilios Liolios, Fotini Papadopoulou, Maria-Anna Rodopoulou, Dimitrios Kampelopoulos, Kostas Siozios, and Chrysoula Tananaki. Decoding the behavior of a queenless colony using sound signals. *Biology*, 12(11):1392, 2023. doi: 10.3390/biology12111392.
- Yun Kang, Krystal Blanco, Talia Davis, Ying Wang, and Gloria DeGrandi-Hoffman. Disease dynamics of honeybees with Varroa destructor as parasite and virus vector. *Mathematical Biosciences*, 275:71–92, 2016. doi: 10.1016/j.mbs.2016.02.012.
- David S. Khoury, Mary R. Myerscough, and Andrew B. Barron. A quantitative model of honey bee colony population dynamics. *PLOS ONE*, 6(4):e18491, 2011. doi: 10.1371/journal.pone.0018491.
- David S. Khoury, Andrew B. Barron, and Mary R. Myerscough. Modelling food and population dynamics in honey bee colonies. *PLOS ONE*, 8(5):e59084, 2013. doi: 10.1371/journal.pone.0059084.
- Alexandra-Maria Klein, Bernard E. Vaissière, James H. Cane, Ingolf Steffan-Dewenter, Saul A. Cunningham, Claire Kremen, and Teja Tscharntke. Importance of pollinators in changing landscapes for world crops. *Proceedings of the Royal Society B*, 274(1608):303–313, 2007. doi: 10.1098/rspb.2006.3721.
- Christopher M. Kribs-Zaleta and Christopher Mitchell. Modeling colony collapse disorder in honeybees as a contagion. *Mathematical Biosciences and Engineering*, 11(6):1275–1294, 2014. doi: 10.3934/mbe.2014.11.1275.
- A Carmen Kuan, Gloria DeGrandi-Hoffman, Robert J Curry, Kristina V Garber, Andrew R Kanarek, Marcia N Snyder, Kurt L Wolfe, and S Thomas Purucker. Sensitivity analyses for simulating pesticide impacts on honey bee colonies. *Ecological Modelling*, 376:15–27, 2018.

- Suzanne Lenhart and John T. Workman. *Optimal Control Applied to Biological Models*. Chapman and Hall/CRC, Boca Raton, FL, 2007.
- Kevin Li, Heather Grab, Mark A. Buckner, Lily Houtman, Melanie Kammerer, Dave McLaughlin, Anaís Ostroski, Margaret R. Douglas, Sarah C. Goslee, Vikas Khanna, Eric V. Lonsdorf, Anthony C. Robinson, and Christina M. Grozinger. Beescape and BeeSpatial: making multi-scale environmental indicators of bee habitat accessible across audiences. *Environmental and Sustainability Indicators*, 2026. doi: 10.1016/j.indic.2026.100849. Volume and article number require verification.
- Eric Lonsdorf, Claire Kremen, Taylor Ricketts, Rachael Winfree, Neal Williams, and Sarah Greenleaf. Modelling pollination services across agricultural landscapes. *Annals of Botany*, 103(9):1589–1600, 2009. doi: 10.1093/aob/mcp069.
- Pierre Magal, Glenn F. Webb, and Yixiang Wu. An environmental model of honey bee colony collapse due to pesticide contamination. *Bulletin of Mathematical Biology*, 81(12):4908–4931, 2019. doi: 10.1007/s11538-019-00662-5.
- Pierre Magal, Glenn F. Webb, and Yixiang Wu. A spatial model of honey bee colony collapse due to pesticide contamination of foraging bees. *Journal of Mathematical Biology*, 80:2363–2393, 2020. doi: 10.1007/s00285-020-01498-7.
- Stephen J. Martin. A population model for the ectoparasitic mite *Varroa jacobsoni* in honey bee (*Apis mellifera*) colonies. *Ecological Modelling*, 109(3):267–281, 1998. doi: 10.1016/S0304-3800(98)00059-3.
- Komi Messan, Gloria DeGrandi-Hoffman, Carlos Castillo-Chavez, and Yun Kang. Migration effects on population dynamics of the honeybee–mite interactions. *Mathematical Modelling of Natural Phenomena*, 12(2):84–115, 2017. doi: 10.1051/mmnp/201712206.
- Komi Messan, Marisabel Rodriguez Messan, Jun Chen, Gloria DeGrandi-Hoffman, and Yun Kang. Population dynamics of Varroa mite and honeybee: effects of parasitism with age structure and seasonality. *Ecological Modelling*, 440:109359, 2021. doi: 10.1016/j.ecolmodel.2020.109359.
- Etienne Minaud, François Rebaudo, Giulia Mainardi, Philippos Vardakas, Fani Hatjina, Ingolf Steffan-Dewenter, and Fabrice Requier. Temperature in overwintering honey bee colonies reveals brood status and predicts colony mortality. *Ecological Indicators*, 169:112918, 2024. doi: 10.1016/j.ecolind.2024.112918. Volume and page numbers require verification.
- Jeffrey M Minucci, Robert Curry, Gloria DeGrandi-Hoffman, Cameron Douglass, Kris Garber, and S Thomas Purucker. Inferring pesticide toxicity to honey bees from a field-based feeding study using a colony model and bayesian inference. *Ecological Applications*, 31(8):e02442, 2021.
- Elynor Moore, Raquel T. de Sousa, Stella Felsinger, Jonathan A. Arnesen, Jane D. Dyekjær, Dudley I. Farman, Rui F. S. Gonçalves, Philip C. Stevenson, Irina Borodina, and Geraldine A. Wright. Engineered yeast provides rare but essential pollen sterols for honeybees. *Nature*, 646(8084):365–371, 2025. doi: 10.1038/s41586-025-09431-y.
- Roger A. Morse and Nicholas W. Calderone. The value of honey bees as pollinators of U.S. crops in 2000. *Bee Culture*, 128(3):1–15, 2000.
- Christian I. Narcia-Macias, Joselito Guardado, Jocell Rodriguez, Joanne Rampersad-Ammons, Erik Enriquez, and Dong-Chul Kim. IntelliBeeHive: an automated honey bee, pollen, and Varroa destructor monitoring system. *arXiv preprint arXiv:2309.08955*, 2023.

- Marcio R. Nunes, Kristen S. Veum, Paul A. Parker, Scott H. Holan, Douglas L. Karlen, Joseph P. Amsili, Harold M. van Es, Skye A. Wills, Cathy A. Seybold, and Thomas B. Moorman. The soil health assessment protocol and evaluation applied to soil organic carbon. *Soil Science Society of America Journal*, 85(4):1196–1213, 2021. doi: 10.1002/saj2.20244.
- A. J. Osorio Rodríguez and H. D. Toro Zapata. Mathematical model for the spread of nosemosis in a honeybee population considering the effects of agrochemicals and control measures. *Theoretical Ecology*, 2025. doi: 10.1007/s12080-025-00632-9.
- Delphine Panziera, F. E. Cousseau, Isobel Grindrod, et al. Liquid and solid matrix formic acid treatment comparison against Varroa mites in honey bee colonies. *Journal of Apicultural Research*, 2023. doi: 10.1080/00218839.2023.2285159. Full author list requires verification.
- Edith M. Patch. Food-plant catalogue of the aphids of the world. *Bulletin of the Maine Agricultural Experiment Station*, 393, 1937. Cited for Patch’s early advocacy on pollinator decline; requires verification.
- J. A. Patiño-Romero et al. An approach to the modeling of honey bee colonies. *Web Ecology*, 22:7–16, 2022. doi: 10.5194/we-22-7-2022. Full author list requires verification.
- Clint J. Perry, Eirik Søvik, Mary R. Myerscough, and Andrew B. Barron. Rapid behavioral maturation accelerates failure of stressed honey bee colonies. *Proceedings of the National Academy of Sciences*, 112(11):3427–3432, 2015. doi: 10.1073/pnas.1422089112.
- Francisco Posada-Flórez, Eugene V. Ryabov, and Yanping Chen. Deadly triangle: honey bees, mites, and viruses. *Frontiers in Bee Science*, 2:1418667, 2024. doi: 10.3389/frbee.2024.1418667. Author list requires verification.
- Thomas G. Preuss, Annika Agatz, Benoit Goussen, Vanessa Roeben, Jack Rumkee, Liubov Zakharova, and Pernille Thorbek. The BEEHAVEecotox model—integrating a mechanistic effect module into the honeybee colony model. *Environmental Toxicology and Chemistry*, 41(11):2870–2882, 2022. doi: 10.1002/etc.5467.
- Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.
- Kirti Rajagopalan, Gloria DeGrandi-Hoffman, Matthew Pruett, Vincent P. Jones, Vanessa Corby-Harris, Julien Pireaud, Robert Curry, Brandon Hopkins, and Tobin D. Northfield. Warmer autumns and winters could reduce honey bee overwintering survival with potential risks for pollination services. *Scientific Reports*, 14:5410, 2024. doi: 10.1038/s41598-024-55327-8.
- Vardayani Ratti, Peter G. Kevan, and Hermann J. Eberl. A mathematical model for population dynamics in honeybee colonies infested with Varroa destructor and the acute bee paralysis virus. *Canadian Applied Mathematics Quarterly*, 21(1):63–93, 2013.
- Vardayani Ratti, Peter G. Kevan, and Hermann J. Eberl. A mathematical model of the honeybee–Varroa destructor–acute bee paralysis virus system with seasonal effects. *Bulletin of Mathematical Biology*, 77(8):1493–1520, 2015. doi: 10.1007/s11538-015-0093-5.
- Vardayani Ratti, Peter G. Kevan, and Hermann J. Eberl. A mathematical model of forager loss in honeybee colonies infested with Varroa destructor and the acute bee paralysis virus. *Bulletin of Mathematical Biology*, 79(6):1218–1253, 2017. doi: 10.1007/s11538-017-0281-6.
- Tomás A. Revilla, Francisco Encinas-Viso, and Michel Loreau. Robustness of mutualistic networks under phenological change and habitat destruction. *Oikos*, 124(1):22–32, 2015. doi: 10.1111/oik.01532.

- Anthony C. Robinson, Xiaoyue Ye, Sterling Scott, Nicholas Treat, and Christina M. Grozinger. Beescape: characterizing user needs for environmental decision support in beekeeping. *Ecological Informatics*, 64:101366, 2021. doi: 10.1016/j.ecoinf.2021.101366. Full author list requires verification.
- Marisabel Rodriguez Messan, Robert E. Page, and Yun Kang. Effects of vitellogenin in age polyethism and population dynamics of honeybees. *Ecological Modelling*, 388:88–107, 2018. doi: 10.1016/j.ecolmodel.2018.09.011.
- Jhonatan O. Rodriguez Rincon, Jun Chen, and Yun Kang. Coupled honey bee–plant–pest–toxin dynamics with endogenous pesticide application. Manuscript in preparation; authorship, title, and venue require confirmation, 2026.
- Peter Rosenkranz, Pia Aumeier, and Bettina Ziegelmann. Biology and control of *Varroa destructor*. *Journal of Invertebrate Pathology*, 103:S96–S119, 2010. doi: 10.1016/j.jip.2009.07.016.
- Jack C. O. Rumkee, Matthias A. Becher, Pernille Thorbek, Peter J. Kennedy, and Juliet L. Osborne. Predicting honeybee colony failure: using the BEEHAVE model to simulate colony responses to pesticides. *Environmental Science & Technology*, 49(21):12879–12887, 2015. doi: 10.1021/acs.est.5b03593.
- Maj Rundlöf, Georg K. S. Andersson, Riccardo Bommarco, Ingemar Fries, Veronica Hederström, Lina Herbertsson, Ove Jonsson, Björn K. Klatt, Thorsten R. Pedersen, Jessica Yourstone, and Henrik G. Smith. Seed coating with a neonicotinoid insecticide negatively affects wild bees. *Nature*, 521(7550): 77–80, 2015. doi: 10.1038/nature14420.
- Sean Russell, Andrew B. Barron, and David Harris. Dynamic modelling of honey bee (*Apis mellifera*) colony growth and failure. *Ecological Modelling*, 265:158–169, 2013. doi: 10.1016/j.ecolmodel.2013.06.005.
- Stenford Ruvinga, Gordon Hunter, Olga Duran, and Jean-Christophe Nebel. Identifying queenlessness in honeybee hives from audio signals using machine learning. *Electronics*, 12(7):1627, 2023. doi: 10.3390/electronics12071627.
- Amer M. Salman, Mohd Hafiz Mohd, A. K. K. Almansoori, et al. Nonlinear dynamics of *Nosema ceranae* and the fragile resilience of honeybee colonies under environmental strain. *Scientific Reports*, 2026. Full author list and volume require verification.
- Marten Scheffer, Stephen R. Carpenter, Timothy M. Lenton, Jordi Bascompte, William Brock, Vasilis Dakos, Johan van de Koppel, Ingrid A. van de Leemput, Simon A. Levin, Egbert H. van Nes, Mercedes Pascual, and John Vandermeer. Anticipating critical transitions. *Science*, 338(6105):344–348, 2012. doi: 10.1126/science.1225244.
- Amelie Schmolke, Farah Abi-Akar, and Silvia Hinarejos. Honey bee colony-level exposure and effects in realistic landscapes: An application of beehave simulating clothianidin residues in corn pollen. *Environmental Toxicology and Chemistry*, 38(2):423–435, 2019.
- Ines Schödl, Richard Odemer, Matthias A. Becher, Stefan Berg, Christoph Otten, Volker Grimm, and Jürgen Groeneveld. Simulation of *Varroa* mite control in honey bee colonies without synthetic acaricides: demonstration of good beekeeping practice for Germany in the BEEHAVE model. *Ecology and Evolution*, 12(11):e9456, 2022. doi: 10.1002/ece3.9456.
- Thomas D. Seeley. *Honeybee Democracy*. Princeton University Press, Princeton, NJ, 2010.
- Abdulhamid Ssenoga, Hasifa Ddumba, and Joseph Y. T. Mugisha. A mathematical analysis and evaluation of *Varroa* mite control methods for a mite-infested honey bee colony. *arXiv preprint arXiv:2205.12732*, 2022.

- Nathalie Steinhauer, Dennis vanEngelsdorp, and Claude Saegerman. Prioritizing changes in management practices associated with reduced winter honey bee colony losses for US beekeepers. *Science of the Total Environment*, 753:141629, 2021. doi: 10.1016/j.scitotenv.2020.141629.
- David J. T. Sumpter and Stephen J. Martin. The dynamics of virus epidemics in Varroa-infested honey bee colonies. *Journal of Animal Ecology*, 73(1):51–63, 2004. doi: 10.1111/j.1365-2656.2004.00776.x.
- Matthew Switanek, Karl Crailsheim, Heimo Truhetz, and Robert Brodschneider. Modelling seasonal effects of temperature and precipitation on honey bee winter mortality in a temperate climate. *Science of the Total Environment*, 579:1581–1587, 2017. doi: 10.1016/j.scitotenv.2016.11.178.
- Horst R. Thieme. Persistence under relaxed point-dissipativity (with application to an endemic model). *SIAM Journal on Mathematical Analysis*, 24(2):407–435, 1993. doi: 10.1137/0524026.
- Horst R. Thieme, Thanate Dhirasakdanon, Zhun Han, and Roberto Trevino. Species decline and extinction: synergy of infectious disease and Allee effect? *Journal of Biological Dynamics*, 3(2–3):305–323, 2009. doi: 10.1080/17513750802376313.
- N. Tsvetkov, O. Samson-Robert, K. Sood, H. S. Patel, D. A. Malena, P. H. Gajiwala, P. Maciukiewicz, V. Fournier, and A. Zayed. Chronic exposure to neonicotinoids reduces honey bee health near corn crops. *Science*, 356(6345):1395–1397, 2017. doi: 10.1126/science.aam7470.
- A. Umara, S. Musa, and A. A. Agada. Analysis of varroosis model in honeybee colony with interventions. *African Scientific Reports*, 2:98, 2023. Bibliographic details require verification.
- Dennis vanEngelsdorp and Marina D. Meixner. A historical review of managed honey bee populations in Europe and the United States and the factors that may affect them. *Journal of Invertebrate Pathology*, 103:S80–S95, 2010. doi: 10.1016/j.jip.2009.06.011.
- Dennis vanEngelsdorp, Jerry Hayes, Robyn M. Underwood, and Jeffery Pettis. A survey of honey bee colony losses in the U.S., fall 2007 to spring 2008. *PLOS ONE*, 3(12):e4071, 2008. doi: 10.1371/journal.pone.0004071.
- Yuanshi Wang and Hong Wu. A mutualism–competition model characterizing competitors with mutualism at low density. *Mathematical and Computer Modelling*, 53(9–10):1654–1663, 2011. doi: 10.1016/j.mcm.2010.12.033.
- Penelope R. Whitehorn, Stephanie O’Connor, Felix L. Wackers, and Dave Goulson. Neonicotinoid pesticide reduces bumble bee colony growth and queen production. *Science*, 336(6079):351–352, 2012. doi: 10.1126/science.1215025.
- Lena Wilfert, Gemma Long, Helen C. Leggett, Paul Schmid-Hempel, Roger Butlin, Stephen J. M. Martin, and Michael Boots. Deformed wing virus is a recent global epidemic in honeybees driven by Varroa mites. *Science*, 351(6273):594–597, 2016. doi: 10.1126/science.aac9976.
- D. Wilkinson and G. C. Smith. A model of the mite parasite, Varroa destructor, on honeybees (*Apis mellifera*) to investigate parameters important to mite population growth. *Ecological Modelling*, 148(3):263–275, 2002. doi: 10.1016/S0304-3800(01)00440-9.
- Sally M. Williamson and Geraldine A. Wright. Exposure to multiple cholinergic pesticides impairs olfactory learning and memory in honeybees. *Journal of Experimental Biology*, 216(10):1799–1807, 2013. doi: 10.1242/jeb.083931.
- Ben A. Woodcock, James M. Bullock, Richard F. Shore, Matthew S. Heard, M. Gloria Pereira, John Redhead, Lucy Ridding, Hannah Dean, Darren Sleep, Peter Henrys, et al. Country-specific effects of neonicotinoid pesticides on honey bees and wild bees. *Science*, 356(6345):1393–1395, 2017. doi: 10.1126/science.aaa1190.

Zainab Zeaiter and Mary R. Myerscough. Poor hive thermoregulation produces an Allee effect and leads to colony collapse. *Journal of Theoretical Biology*, 503:110361, 2020. doi: 10.1016/j.jtbi.2020.110361.

A Structured inventory of post-2022 contributions

The tables below inventory contributions since 2022 across four of the five thematic pillars, compiled during workshop follow-up. They are intended for high recall rather than strict selectivity: mathematical models, relevant reviews, and empirical datasets useful for parameterization are all included, and entries appear in more than one table where the contribution spans categories.

Verification status. Entries drawn from the working inventory carry a verification flag in the source spreadsheet. Where authorship or bibliographic details could not be confirmed at the time of writing, this is noted in references .bib and those entries must be verified before submission.

Table 5: Persistence and collapse: post-2022 contributions.

Reference	Focus	Method	Contribution
Ibrahim and Dénes [2022]	<i>Varroa destructor</i> –virus, seasonality	Seasonal ODE	Parallel spread of mites and vectored virus with time-periodic parameters; includes infected foragers
Chen et al. [2023]	Seasonality and parasitism	Non-autonomous nonlinear ODE	Effects of seasonality and parasitism on colony size and stability, with bifurcation analysis
Ssenoga et al. [2022]	<i>Varroa destructor</i> management	ODE, reproduction numbers	Mite-free and mite reproduction numbers; persistence and extinction conditions
[Authors to verify] [2025]	Threshold dynamics	ODE threshold model	Explicit basic reproduction number determining colony survival or collapse under general eclosion and recruitment rates
Alharbi [2024]	Disease and seasonality	Non-autonomous model	Two-hive disease model with reproduction numbers in autonomous and non-autonomous settings
Rajagopalan et al. [2024]	Climate, overwintering	VarroaPop simulation	Warmer autumns extend flight, skew overwintering age structure, and raise spring failure risk; cold storage as mitigation
Patiño-Romero et al. [2022]	Colony dynamics, honey yield	Mathematical/statistical	Adult and immature bee dynamics with honey production
<i>Observations.</i> Seasonality, parasitism, overwintering, and threshold dynamics recur. The category overlaps substantially with disease and optimization. No post-2022 entry analyses more than two stressor categories as a coupled system.			

B Notes on sources and reference verification

This manuscript consolidates material from the AIM workshop final report [AIM Workshop Participants, 2026], two prior manuscript drafts, thematic literature summaries on agrochemicals and climate, a disease modelling summary, and a working spreadsheet inventory. During consolidation the reference list was cross-checked, and the following corrections were applied relative to the earlier drafts.

- The sterol supplementation study previously cited as “Wright et al. (2025)” is correctly Moore et al. [2025], *Nature* **646**(8084), 365–371.
- Ratti et al. [2015] and Ratti et al. [2017] are distinct papers and were previously conflated: the 2015 paper concerns the seasonal bee–*Varroa destructor*–ABPV system, the 2017 paper concerns forager loss.
- The Bernardi–Venturino model appeared in *Heliyon* [Bernardi and Venturino, 2016], not in *Mathematical Biosciences*.
- The Beescape reference is Li et al. [2026] in *Environmental and Sustainability Indicators*, which also covers BeeSpatial.
- A colony-loss surveillance citation attributed to “Steinhauer et al. (2021), *Apidologie*” could not be located; Bruckner et al. [2023] and Steinhauer et al. [2021] are substituted.

Table 6: Disease, mites, and viruses: post-2022 contributions.

Reference	Focus	Method	Contribution
Ibrahim and Dénes [2022]	<i>Varroa destructor</i> –virus, seasonality	Seasonal ODE	Mite and vectored virus spread with periodic parameters
Ssenoga et al. [2022]	<i>Varroa destructor</i> infestation	ODE, reproduction numbers	Mite-infested colony model with control strategy evaluation
Chen et al. [2023]	Parasitism	Non-autonomous ODE	Parasitism reduces colony size and may destabilize dynamics under seasonal laying
Alharbi [2024]	Disease, seasonality	Non-autonomous disease model	Two hives sharing a common space; $\mathcal{R}_0$ for both settings
Umara et al. [2023]	Varroosis with interventions	Compartmental model	Combined treatment and biocontrol strategies
Osorio Rodríguez and Toro Zapata [2025]	Nosemosis, agrochemicals	Compartmental model with control	Links pathogen transmission to an environmental stressor; among the clearest recent multi-stressor disease models
El Hajji et al. [2026]	Dual <i>Nosema</i> , two colonies	DDE / ODE analysis	<i>N. apis</i> and <i>N. ceranae</i> across interacting colonies with distributed delays and environmental pathogen compartment
Salman et al. [2026]	<i>N. ceranae</i> , environmental strain	Generalized SIR	Limited treatment capacity, management constraints, transient suppression
Posada-Flórez et al. [2024]	<i>Varroa destructor</i> –virus complex	Review	Synthesis of mite-vectored virus emergence, virulence evolution, and multi-stressor framing; identifies the coinfection gap
Abban et al. [2024]	<i>Varroa destructor</i> and <i>Nosema</i> prevalence	Empirical surveillance	US-wide prevalence 2015–2022; parameterization and burden data
<i>Observations.</i> New mathematical work since 2022 includes <i>Varroa destructor</i>–seasonality models, seasonal disease models, varroosis intervention models, and single- and dual-species <i>Nosema</i> models. Explicit mathematical coinfection models remain rare, though empirical coinfection and microbiome studies increasingly motivate them.			

Table 7: Optimization and management: post-2022 contributions.

Reference	Focus	Method	Contribution
Ssenoga et al. [2022]	<i>Varroa destructor</i> management	ODE, threshold analysis	Compares chemical, mechanical, and traditional control including queen caging and replacement
Schödl et al. [2022]	Management evaluation	BEEHAVE simulation	Drone brood removal, formic and oxalic acid, and combinations, without synthetic acaricides
Panziera et al. [2023]	Intervention comparison	Experimental evaluation	Liquid versus solid matrix formic acid delivery
<i>Observations.</i> Very few post-2022 papers perform formal optimization; most <i>evaluate</i> interventions rather than optimizing over a control space. <i>Varroa destructor</i> management dominates. Colony splitting, queen replacement, supplemental feeding, overwintering strategy, and multi-objective management are all substantially underrepresented. Additional entries in the working inventory require bibliographic verification and have been omitted here.			

Table 8: Early warning, detection, and monitoring: post-2022 contributions.

Reference	Focus	Method	Contribution
Hall et al. [2023]	<i>Varroa destructor</i> detection	ML on vibration signals	Non-invasive detection from brood-comb vibrational gait signatures
Divasón et al. [2023]	<i>Varroa destructor</i> detection	Deep learning, vision	Image-based mite detection relevant to treatment timing
Duma et al. [2024]	<i>Varroa destructor</i> detection	Hyperspectral imagery	Spectral-band detection of mites on bees
Narcia-Macias et al. [2023]	Hive activity, <i>Varroa destructor</i>	Object detection (YOLO family)	Integrated automated monitoring of bees, pollen, and mites at the entrance
Ruvinga et al. [2023]	Queenlessness	ML on audio	Detection of queenless state from acoustic signals
Kanelis et al. [2023]	Queenlessness	Acoustic signal analysis	Characterizes sound changes following queen removal
Minaud et al. [2024]	Overwintering mortality	Temperature time series	Distinguishes surviving from failing colonies and predicts mortality
Degenfellner and Templ [2024]	Colony condition	ML, trend analysis	Infers colony condition from hive weight and environmental parameters
<i>Observations.</i> This category is larger than optimization and overlaps heavily with machine learning and precision apiculture. Most entries perform <i>detection</i> of a current state rather than <i>prediction</i> of a future one; the entries closest to genuine early warning are those exploiting time series structure. Several further entries in the working inventory require bibliographic verification and are omitted here.			

- Citations previously listed as “Yamada et al. (2019)”, “Nunes-Parker et al. (2021)”, and an in-preparation trajectory-analysis manuscript could not be verified and have been removed. The soil health assessment work referenced at the workshop is Nunes et al. [2021].

Entries in `references.bib` carrying a `note` field indicating that details require verification should be confirmed against publisher records before submission. These are principally recent or non-indexed items: Umara et al. [2023], El Hajji et al. [2026], Salman et al. [2026], [Authors to verify] [2025], Patiño-Romero et al. [2022], Panziera et al. [2023], Duma et al. [2024], Minaud et al. [2024], Robinson et al. [2021], Bromenshenk et al. [2009], Posada-Flórez et al. [2024], and Rodríguez Rincon et al. [2026].

Content requiring author confirmation. Several substantive claims in this manuscript derive from workshop discussion or from work in progress rather than from published results, and are identified as such in the text. These include: the composite stress score Σ and its proposed operational definition (Section 4); the parameter values and descriptive statistics of the in-hive sensor dataset (Section 9); the structural results attributed to Rodríguez Rincon et al. [2026] (Section 6); and the cold-storage field comparison referenced in Section 6. Each should be either published, cited to a preprint, or attributed as personal communication before submission.